**[ACCEPTED VERSION]**

# Unequal Risks, Unequal Responses: Differential Effects of Emphasizing Race vs. Class Disparities in News Stories about Climate Impacts


## Abstract

We investigated how news articles discussing unequal impacts of climate change due to race vs. class influence beliefs and support for action. We conducted two online survey experiments focusing on flooding (Study 1) and heat waves (Study 2). In Study 1, White participants and those with high levels of symbolic racism had lower levels of belief and support when exposed to the race-focused condition, although these interactions were not as consistent in Study 2. Results suggest that focusing on class-based climate disparities may be less susceptible to backfire effects than focusing on race-based disparities.



Keywords: *Climate Change, Symbolic Racism, Climate Justice, Quantitative Methodology*

**Unequal Risks, Unequal Responses: Differential Effects of Emphasizing Race vs. Class Disparities in News Stories about Climate Impacts**

Scientists broadly agree that climate change disproportionately impacts low-income communities and communities of color ("IPCC, 2023: Summary for Policymakers," 2023). Yet recent research has found that the U.S. public is not widely aware of these inequities (Bugden, 2024; Schuldt & Pearson, 2023) and, as a result, may not support policies and programs intended to address these vulnerabilities (Bergquist et al., 2020). Moreover, Americans are more likely to acknowledge class-based than race-based vulnerabilities to climate change (Bugden, 2024; Schuldt & Pearson, 2023). It is therefore possible that individuals will be less receptive to information about unequal impacts of climate change when it highlights racial rather than class divides, particularly if they are not part of the at-risk racial group or if they hold racist attitudes. While several studies have examined how the media discusses unequal impacts of climate change (Das, 2020; Dreher & Voyer, 2015; Russell et al., 2023; A. Schmidt & Schäfer, 2015), the two studies presented here are the first we are aware of to examine how exposure to news stories that emphasize the unequal impacts of climate change based on race vs. class may influence public perceptions. Perceptions of these disparities may further shape support for public policies designed to address climate inequities (Schuldt & Pearson, 2023). Drawing from literature on social identity (Mackay et al., 2021) and symbolic racism (Sears & Henry, 2005), the studies are also the first we are aware of to investigate how effects of climate inequality narratives may be moderated by the message recipient's race and symbolic racism. The studies examine how climate inequality narratives may shift beliefs in unequal impacts, support for impacted communities, and policy support in the context of two different climate risks: flooding (Study 1) and heat waves (Study 2).

## Literature Review

The negative impacts of climate change are broad and affect multiple aspects of society, including weather patterns, air quality, the economy, human health, ecosystems and biodiversity ("IPCC, 2023: Summary for Policymakers," 2023). While climate change is affecting almost all aspects of life, these impacts are not evenly distributed – climate change risks such as flooding and heat waves are more likely to impact communities with lower socioeconomic status and higher concentrations of minoritized individuals (Schlosberg & Collins, 2014). In recent years, advocacy groups and government organizations have often noted the importance of protecting vulnerable populations as justification for action to mitigate and adapt to climate change (Schlosberg & Collins, 2014; US EPA, 2022).

Despite broad scientific agreement about unequal climate impacts, there is not a broad public belief in these unequal effects. Recent survey data finds that only 37-44% of U.S. respondents believe that climate change has unequal impacts, and only 22% of U.S. respondents believe that some racial groups are more vulnerable to climate impacts than others (Schuldt & Pearson, 2023). We note that while public understanding is limited, media coverage of climate change has included stories focused on how climate change may have larger impacts on specific groups (Russell et al., 2023). Russel et al. (2023) found that journalists have a strong interest in telling stories highlighting the unequal impacts of climate change and how these unequal impacts are linked to structural racism and racial injustice. While prior research has investigated patterns in media coverage of climate justice and unequal climate impacts (Das, 2020; Dreher & Voyer, 2015; Russell et al., 2023; A. Schmidt & Schäfer, 2015), few studies have investigated how the public is likely to respond to such stories.

Several conceptual articles suggest that communicating about unequal impacts may have unintended effects. Tsai and Pearson (2022) state that climate change communication practitioners often assume that "conveying climate-related inequity enhances support for efforts to reduce inequity among both advantaged and disadvantaged groups" (p. 525), but note that this assumption is problematic and that communicating about inequities instead may increase polarization and potentially cause both advantaged and disadvantaged groups to be less likely to take action. Pearson et al. (2024) raise the concern that increased knowledge about unequal climate impacts may paradoxically increase stigmatization of more vulnerable groups and cause individuals to support policies that reinforce inequities rather than reduce them. Swim and Bloodhart (2018) argue that individuals may be predisposed to minimize or not recognize unequal impacts of climate change due to a number of psychological factors, including motivated cognition, lack of efficacy, stereotypes of activists and impacted communities, and climate misinformation. While these conceptual articles raise potential unintended consequences of communicating about the unequal impacts of climate change, empirical testing of these claims has been limited. One of the few studies to empirically investigate this area does suggest that narratives about climate inequities can be polarizing (Whitmarsh & Corner, 2017).

Given the lack of experimental research testing narratives about unequal climate impacts, conceptual work on the potential impact of such climate messages has predominantly drawn from research testing the effects of communicating about social disparities in health contexts (Pearson et al., 2024; Tsai & Pearson, 2022). The health disparities communication literature generally finds that information about economic and class-based inequities sometimes leads to broader public support for intervention, while race-based narratives tend to decrease support and increase polarization (Liu & Niederdeppe, 2024). For example, Rigby et al. (2009) found that

providing respondents with information about racial health inequalities lowered perceptions of social causes of the inequities and support for government intervention compared to providing information about economic health disparities. Schmidt et al. (2022) found that individuals were more supportive of equity-based Covid-19 vaccine distributions when provided information about "economically disadvantaged neighborhoods" being harder hit by Covid-19 compared to information about minoritized communities being more impacted.

Climate change has several characteristics that differentiate it from many health issues and make it a worthwhile unique substantive domain for inquiry. For example, climate change is considered a "super wicked" problem, in which individuals have limited agency to directly mitigate the issue, and efforts to address climate change require invoking immediate costs to prevent severe negative consequences in the future (Lazarus, 2008; Levin et al., 2012). In addition, environmental issues in general, and climate change specifically, are the two most politically polarizing issues in the U.S. today (Pew Research Center, 2024); more crystalized views about climate change may make the public less sensitive to different message strategies for discussing unequal impacts.

Thus, while the present research builds on previous work in the health communication literature, it also offers a novel contribution by examining how the public responds to inequality narratives in the context of climate change. Understanding the effects of such narratives holds important implications for increasing awareness of climate vulnerabilities, reducing racial and social inequities in climate risks, and encouraging support for related policies (e.g., Pearson et al. 2024; Schuldt & Pearson, 2023). As such, we focus on how these narratives might shape beliefs in unequal impacts, support for impacted communities, and support for policies that address climate risks.

In the present study, we compare how the public responds to information about the increased risk of climate impacts in general and increased risk that has unequal impacts due to race vs. class. For race, we focus on risks to Black communities, which have been identified as having a high level of vulnerability to climate impacts (Schlosberg & Collins, 2014; United States Environmental Protection Agency, 2021; U.S. Department of Health and Human Services, 2022). Because different language can be used to describe lower socioeconomic status (SES) groups, we include two class conditions, one using the term "poor" and the other using the term "working class." Formally stated, we ask the research question:

*RQ1*: How does a news story about climate change–driven extreme weather, either with (a) no group emphasis or with an emphasis on disproportionate impacts to (b) Black communities, (c) poor communities, or (d) working-class communities, influence beliefs about unequal impacts, support for impacted communities, and support for climate policies?

### *Moderating Effects of Symbolic Racism and Race*

The studies presented here offer an additional theoretical contribution by being the first we are aware of to investigate how symbolic racism beliefs and the race of message recipients may moderate the effects of messages about the unequal impacts of climate change. We discuss these moderating factors below.

**Symbolic Racism.** Symbolic racism is a coherent belief system rooted in the ideas that racial discrimination is no longer a problem and that Blacks do not deserve special treatment in society (Sears & Henry, 2003, 2005). While symbolic racism is correlated with "old-fashioned racism" (Sears & Henry, 2005, p. 96) and political ideology, it has unique effects as a theoretical construct and is an important conceptual measure of modern racism when individuals may be less likely to agree with measures of explicit racism due to social desirability (Sears & Henry,

2003, 2005). Several recent studies have found that individuals with higher levels of symbolic racism have lower levels of environmental concern and are less likely to take action to address environmental issues (Benegal, 2018; Chanin, 2018; Dietz et al., 2018). Harell and Liebermann (2021), while not measuring symbolic racism specifically, found that general sentiment towards Black communities may moderate responses to health disparities information; this suggests that symbolic racism may be an important moderator that can interact with class and race-based information about unequal climate impacts. Given that symbolic racism is measured, in part, by agreement with statements that Black communities need to work harder to achieve higher status in society and should not be given special treatment (Henry & Sears, 2002; Sears & Henry, 2005), we expect that individuals who have beliefs aligned with symbolic racism will be especially skeptical of claims that climate change is having an unequal impact on Black communities. Formally stated:

*H1:* Symbolic racism will have a negative relationship with belief in unequal impacts, support for impacted communities, and policy support to address climate change.

*H2:* Symbolic racism will moderate message effects such that exposure to information about unequal climate impacts on Black communities, compared to (a) control messages or (b) messages focusing on class-based vulnerabilities, will have a more pronounced negative effect on belief in unequal impacts, support for impacted communities, and policy support among individuals who have higher levels of symbolic racism.

**Race.** In addition to symbolic racism beliefs, the race of message recipients may also moderate how messages are received. Recent work has called for more research on how demographic factors such as race may shape environmental perceptions and responses to climate change messages (Raimi et al., 2024). Previous work has found that racial minorities and those

most vulnerable to climate change in the U.S. have heightened environmental concerns (Lazri & Konisky, 2019; Mackay et al., 2021; Medina et al., 2019; Pearson et al., 2018; Pearson & Schuldt, 2015) and are more supportive of climate justice (Carman et al., 2025). In addition, social identity theory (Hart & Nisbet, 2012; Hornsey, 2008) as well as the social identity approach (Mackay et al., 2021) suggest that group identification impacts risk perceptions and how messages are likely to be received. Moreover, exposure to information about negative climate change impacts can increase ethnocentrism and outgroup derogation (Shanaah et al., 2024; Uhl et al., 2018). Individuals are predisposed to protect in-groups they belong to (Mackay et al., 2021), and those who identify with more vulnerable populations are likely to be more supportive of mitigative and adaptive efforts to address climate change. In contrast, individuals who identify with higher-status groups may resist actions seen as disrupting the status quo. Therefore, we hypothesize that:

*H3:* Black participants will have higher beliefs in unequal impacts, support for impacted communities, and policy support compared to White participants.

*H4:* Race will moderate the response to messages about unequal impacts of climate change. We expect that compared to (a) control messages or (b) messages focusing on class-based vulnerabilities, messages focusing on the vulnerabilities of Black communities may increase Blacks' beliefs in unequal impacts, support for impacted communities, and policy support while decreasing these variables among Whites.

## Overview of Studies

The literature discussed above suggests that messages about unequal impacts of climate change that focus on race vs. class may have different effects on recipients' beliefs and support for action, and that these effects may vary depending on their level of symbolic racism and their

race; however, to date, previous research has not examined these three factors in combination with each other. To address this research gap, we examined how constructed news stories about unequal impacts of climate change may influence public perceptions, as moderated by symbolic racism and race, respectively. We focused on two impacts of climate change: flooding and heat waves. Climate change is increasing the risks of flooding and heat waves, both of which disproportionately impact vulnerable populations such as lower-SES and minoritized communities (Schlosberg & Collins, 2014; United States Environmental Protection Agency, 2021; U.S. Department of Health and Human Services, 2022); however, risk perceptions associated with these two impacts may be different. For example, flooding tends to create more localized risk perceptions than heat waves (Lohmann & Kontoleon, 2023). By examining both types of impacts, we are able to offer more generalizable findings that are less at risk to a case confound occurring from unique characteristics of a specific impact type. Thus, we conducted two studies: Study 1 focused on the risk of flooding and Study 2 focused on the risk of heat waves.

In both studies, the dependent variables include beliefs in unequal impacts, support for impacted communities, and support for policies that address climate risks. Because individuals may respond differently to questions focused on climate change compared to questions focused on specific risks like flooding (Hart et al., 2023), we included separate measurement around specific impacts (flooding in Study 1, heat waves in Study 2) and climate change for each dependent variable. While recent research has been critical of a deficit-based, "just the facts" approach to science communication (Nisbet, 2009), presenting evidence and establishing a common ground for the reality of climate change is still considered an important aspect of communication efforts (Garrett, 2017). Thus, we measure how beliefs that climate change and

flooding (Study 1) / heat waves (Study 2) have unequal impacts may change in response to whether information emphasizes unequal climate impacts associated with either race or class. Given that climate change is an issue requiring large-scale social action (Hart et al., 2023; Lazarus, 2008; Levin et al., 2012; Thier & Lin, 2022), we also measure support for two types of large-scale social response: (1) support for communities impacted by climate change and flooding / heat waves, and (2) support for policies to address climate change and flooding / heat waves.

## Methodology

We used G*Power 3.1.9.6 to estimate the sample size needed to test the research questions and hypotheses under investigation. For a 5-condition ANOVA with $\alpha = 0.05$ and power = 0.8, for small effects (Cohen's $f = 0.1$), a total N of at least 1,200 is needed, while for medium effects (Cohen's $f = 0.25$) a total N of at least 200 is needed. For interactions, we predicted a reverse-slope effect for race by condition (*H*4) and a fully attenuated interaction for symbolic racism by condition (*H*2); Giner-Sorola (2018) offers the guidance that for reverse-slope interactions, studies should use a sample size of at least twice the total *N* as used for main effects, and for fully-attenuated interactions scholars should use at least four times the total *N* as used for main effects. Thus, for main effects our studies are well powered to detect small effects, but for the two-way interactions under investigation, the studies are only sufficiently powered to detect medium effects.

Participants were recruited from Forthright, an online sampling platform. The samples for both studies included a U.S. nationally diverse sample that roughly matched census characteristics, with an additional oversample of 375 Black respondents to provide a large enough Black sample to compare responses of Black participants with White participants. The

Study 1 sample ($N$ = 1,434) had the following characteristics: 53.6% White only, 35.1% Black only, 11.2% mixed race or race other than Black or White; age range of 18-87 ($M$ = 46.2, $SD$ = 16.31); 47.0% male, 52.0% female, 1.0% non-binary / other. The Study 2 sample ($N$ = 1,421) had the following characteristics: 55.9% White only, 34.9% Black only, 9.2% mixed race or race other than Black or White; age range of 18-88 ($M$ = 46.56, $SD$ = 16.25); 45.6% male, 52.7% female, 1.7% non-binary / other).

After consenting to participate in the online experiment, participants were randomly assigned to one of five experimental conditions. Participants in the stimulus conditions were asked to read a constructed AP news article that described how climate change is expected to increase the risk of either flooding (Study 1) or heat waves (Study 2) in the U.S. over the next few decades. The news story either included information that identified (1) Black communities, (2) poor communities, (3) working class communities, or (4) did not identify a specific community (referred to as "No Group") as being most at risk from climate change impacts. The rest of the news story was held consistent across the conditions. The final condition was a (5) no message control (referred to as "Control"). Random assignment was successful – there was not a significant difference in Study 1 or Study 2 in the distribution of the race, age, or gender of the participants across the experimental conditions.[1] The full text of the stimuli can be found in the Supplemental Material.

Participants in the stimulus conditions were asked to read the constructed news story about a specific risk of climate change (flooding in Study 1, heat waves in Study 2; the full text of the stimuli can be found in the Supplemental Material). After reading the news story,

[1] Because random assignment was successful, we did not control for demographic variables in the results reported below. We did conduct a robustness check of the analyses including the demographic variables as controls, and the pattern of results was identical to the results reported here.

participants answered questions to measure the following six dependent variables: (1) beliefs in unequal impacts from flooding / heat waves and (2) climate change; (3) support for communities impacted by flooding / heat waves and (4) climate change; and (5) support for policies to address flooding / heat waves and (6) climate change. Participants in the control condition answered identical questions but were not exposed to a news story first. Following the dependent variables, we measured symbolic racism beliefs and race to allow for an investigation of how these factors may moderate differences between message conditions. See Table 1 for measurement text and variable descriptives for all dependent variables and symbolic racism.

For the analyses below, we first used ANOVA to investigate the main effect of the message treatment on the respective dependent variables. We then used OLS regression to investigate how differences between conditions may be moderated by levels of symbolic racism and then probed significant interactions with a Johnson-Neyman analysis to identify regions of significance. Next, we used ANOVA to investigate how participants identifying as Black or White may differentially respond to the variables of interest, and whether the race of the participant may interact with the message condition in predicting the dependent variables (Note: we focused on participants who identified as Black only or White only because identification with other specific racial groups (e.g. Asian) were 4.5% or less of the experimental sample in Study 1 and Study 2, preventing a meaningful analysis of other racial categories).

## Results

### Study 1: Flooding Impacts

#### *Main Effects of Condition*

We first looked at main effects of the conditions to investigate RQ1: how public responses may change in response to news stories emphasizing race vs. class. ANOVA analyses

revealed a significant main effect of message condition on beliefs in unequal impacts of flooding (F[4, 1429] = 10.36, $p < .001$, $\eta_p^2 = .028$) and climate change (F[4, 1428] = 4.88, $p < .001$, $\eta_p^2 = .013$), and support for communities impacted by climate change (F[4, 1428] = 2.83, $p = .023$, $\eta_p^2 = .008$). There were no main effects of condition on any other variables; see Table 2 and Figure 1 for means by condition and ANOVA results for all variables.

Sidak post hoc tests for beliefs in unequal impacts of flooding revealed that participants in the Black condition ($M = 4.23$) had significantly lower beliefs than participants in all other four conditions: Control ($M = 4.65$, $p < .001$), No Group ($M = 4.51$, $p < .001$), Poor ($M = 4.51$, $p < .001$), and Working Class ($M = 4.43$, $p = .034$). Additionally, participants in the Working Class condition had lower beliefs compared to the Control condition ($p = .009$).

Sidak post hoc tests for beliefs in unequal impacts of climate change revealed that participants in the Control condition ($M = 3.67$) had significantly lower beliefs than participants in the Poor ($M = 4.06$, $p = .002$) and Working Class ($M = 3.99$, $p = .019$) conditions. Participants in the Black condition ($M = 3.74$) had significantly lower beliefs than participants in the Poor condition ($p = .030$). There were no significant differences between participants in the No Group condition ($M = 3.85$) and the other conditions, and no other comparisons were significant.

Sidak post hoc tests for support for communities impacted by climate change found that participants in the Control condition ($M = 3.79$) had significantly less support than participants in the Poor condition ($M = 4.07$, $p = .023$). There were no significant differences in comparisons with participants in the No Group ($M = 4.00$), Black ($M = 3.92$), or Working Class ($M = 4.02$) conditions, and no other comparisons were significant.

***Symbolic Racism***

Because symbolic racism is a continuous variable, we used hierarchical regression to analyze interactions between symbolic racism and the conditions. We first entered a block that included all main effects and then entered a second block that included the interaction between dummy coded message conditions and symbolic racism.

We first looked at main effects to test *H1*: Symbolic racism will have a negative relationship with the dependent variables. In support of *H1*, symbolic racism had a significant negative relationship with all dependent variables: belief in unequal impacts of flooding (unstandardized $b = -.142$, $p < .001$) and climate change ($b = -.318$, $p < .001$), support for communities impacted by flooding ($b = -.263$, $p < .001$), support for communities impacted by climate change ($b = -.464$, $p < .001$), support for policies to address flooding ($b = -.183$, $p < .001$), and support for policies to address climate change ($b = -.438$, $p < .001$). The OLS regression table showing the relationships between symbolic racism and all dependent variables can be found in Supplemental Material Table S1.

We next looked to interactions between symbolic racism and conditions to investigate *H2:* symbolic racism will moderate message effects such that exposure to information about unequal climate impacts on Black communities, compared to (a) the control conditions or (b) class focused messages, will have a more pronounced negative effect on the dependent variables for individuals who have higher levels of symbolic racism. Including the interaction terms created a significant $R^2$ change for five of the six dependent variables: belief in unequal impacts of flooding ($\Delta R^2 = .023$, $p < .001$), belief in unequal impacts of climate change ($\Delta R^2 = .018$, $p < .001$), support for communities impacted by flooding ($\Delta R^2 = .011$, $p < .001$), support for communities impacted by climate change ($\Delta R^2 = .006$, $p = .018$), and support for policies to address flooding ($\Delta R^2 = .008$, $p = .011$). There was not a significant $R^2$ change for support for

policies to address climate change when interaction terms were included. Below we discuss the significance of symbolic racism with specific condition comparisons; please see Table 3 for interaction coefficients and Johnson-Neyman results for all comparisons and Figure 2 for a visual representation of all interactions. The full regression tables for all interactions between conditions and symbolic racism can be found in Supplemental Material Tables S2 – S5. These interactions, as described in more detail below, exhibit a general pattern whereby for participants with high levels of symbolic racism, the Black condition resulted in lower levels of beliefs and support relative to some or all of the other message conditions, whereas for participants with low levels of symbolic racism, the Black condition boosted beliefs and support. However, the negative effects of the Black condition among those with higher levels of symbolic racism occurred more frequently than the positive effects among those with lower levels of symbolic racism. Some of the other pairwise comparisons between conditions (e.g., between the No Group and Control conditions) also varied based on symbolic racism, but the patterns were not consistent.

For belief in unequal impacts of flooding, symbolic racism had a significant interaction such that participants with higher levels of symbolic racism had lower beliefs when exposed to the Black condition compared to all other conditions. People with higher levels of symbolic racism also had lower beliefs when exposed to the Poor and Working Class conditions compared to the Control condition. No other interactions were significant.

For belief in unequal impacts of climate change, for participants with higher levels of symbolic racism, exposure to the Black condition compared to the No Group and Working Class conditions lowered beliefs. For participants with lower levels of symbolic racism, exposure to

the Black and Poor conditions increased beliefs compared to the No Group condition. No other interactions were significant.

For support for communities impacted by flooding, among participants with high levels of symbolic racism, the Black condition decreased support compared to the Control, Poor, and Working Class conditions. The No Group condition also decreased support relative to the Control condition among those with high symbolic racism scores. Among those with lower levels of symbolic racism, the Black condition increased support relative to the Control condition. No other interactions were significant.

For support for communities impacted by climate change, among those with high levels of racism, the Black condition decreased support relative to the No Group, Poor, and Working Class conditions. Among those with low levels of racism, the Black condition increased support but only compared to the Control condition. No other interactions were significant.

For support for policies to address flooding, the comparisons were significant only among participants with low levels of racism. For this group, the Black condition increased support relative to the Control and Working Class conditions, and the No Group condition also increased support relative to the Control. No other interactions were significant.

***Race***

We next used ANOVA to analyze *H3:* Black participants will score higher on the dependent variables compared to White participants, and *H4:* race will moderate message effects such that exposure to information about unequal climate impacts on Black communities, compared to (a) the control conditions or (b) class focused messages, may increase Blacks' beliefs in unequal impacts, support for impacted communities, and policy support while decreasing these variables among Whites. For the race comparisons, we only included

participants ($n$ = 1,273) who identified as Black only or White only, as the sample size for other races was not sufficient for meaningful analysis.

The ANOVA revealed that race had a significant association with five of the six dependent variables, such that Whites, compared to Blacks, had lower beliefs in unequal impacts of climate change, support for communities impacted by flooding and climate change, and support for policies to address flooding and climate change. The significant mean differences were as follows: belief in unequal impacts of climate change ($F[1, 1262] = 11.98$, $p < .001$, $\eta_p^2 = .009$; $M_{White} = 3.77$, $M_{Black} = 4.02$), support for communities impacted by flooding ($F[1, 1261] = 55.64$, $p < .001$, $\eta_p^2 = .042$; $M_{White} = 4.10$, $M_{Black} = 4.44$), support for communities impacted by climate change ($F[1, 1262] = 92.49$, $p < .001$, $\eta_p^2 = .068$; $M_{White} = 3.74$, $M_{Black} = 4.32$), support for policies to address flooding ($F[1, 1262] = 19.20$, $p < .001$, $\eta_p^2 = .015$; $M_{White} = 4.14$, $M_{Black} = 4.31$), and support for policies to address climate change ($F[1, 1262] = 29.18$, $p < .001$, $\eta_p^2 = .023$; $M_{White} = 3.85$, $M_{Black} = 4.17$).

The ANOVA found significant interactions between condition and race for four of the six dependent variables: belief in unequal impacts of flooding ($F[4, 1263] = 2.92$, $p = .020$, $\eta_p^2 = .009$), support for communities impacted by flooding ($F[4, 1261] = 4.69$, $p < .001$, $\eta_p^2 = .015$), support for communities impacted by climate change ($F[4, 1262] = 2.59$, $p = .035$, $\eta_p^2 = .008$), and support for policies to address flooding ($F[4, 1262] = 3.21$, $p = .012$, $\eta_p^2 = .010$) (See Figure 3 and Supplemental Material Table S6 for means of all dependent variables by race and condition for Study 1.)

We probed the significant interactions by examining the simple effects of message conditions within each racial group (i.e., Black and White). For the four variables with a significant interaction between race and message condition, the simple effects of message

condition were significant only for White participants: unequal impacts of flooding (F[4, 1263] = 11.89, $p < .001$, $\eta_p^2 = .036$), support for communities impacted by flooding (F[4, 1261] = 6.54, $p < .001$, $\eta_p^2 = .020$), support for communities impacted by climate change (F[4, 1262] = 5.35, $p < .001$, $\eta_p^2 = .017$), and support for policies to address flooding (F[4, 1262] = 3.61, $p = .006$, $\eta_p^2 = .011$). Among Black participants, there were no differences between conditions. We used the Sidak adjustment to examine pairwise comparisons between message conditions among White participants, and we report the significant differences below.

For belief in unequal impacts of flooding, White participants had significantly lower perceptions in the Black condition than in the Control ($p < .001$), No Group ($p < .001$), Poor ($p < .001$) or Working Class ($p < .001$) conditions. No other comparisons were significant.

For support for communities impacted by flooding, White participants exhibited significantly less support in the Black condition than in the Control ($p = .013$), Poor ($p < .001$) and Working Class ($p < .001$) conditions. No other comparisons were significant.

For support for communities impacted by climate change, White participants exhibited significantly less support in the Black condition than in the poor ($p = .011$) and Working Class ($p = .020$) conditions. White participants also exhibited significantly less support in the Control condition than in the Poor ($p = .006$) and Working Class ($p = .012$) conditions.

For support for policies to address flooding, White participants exhibited significantly less support in the Control condition than participants in the Working Class condition ($p = .024$). No other comparisons were significant.

**Study 2: Heat Waves**

Study 2 used the same experimental structure, variables, and analytic approach as Study 1. The only difference was the risk context: The news stimuli in Study 2 focused on how climate

change is increasing the risks of heat waves, whereas the news stimuli in Study 1 focused on flooding risks (see Table 1 for text and descriptive statistics for all variables in Study 2; see Supplemental Material for the full text of the stimuli).

***Main Effects of Condition***

ANOVA analyses revealed a significant main effect of message condition (*RQ1*) on belief in unequal effects of heat waves ($F$[4, 1416] = 7.69, $p < .001$, $h^2_p = .021$), support for communities impacted by heat waves ($F$[4, 1416] = 2.64, $p = .032$, $\eta^2_p = .007$), and support for communities impacted by climate change ($F$[4, 1414] = 3.45, $p = .008$, $\eta^2_p = .010$). There were no main effects of condition on any other variables; see Table 4 and Figure 4 for means by condition and ANOVA results for all variables.

Sidak post hoc tests for beliefs in unequal impacts of heat waves revealed that participants in the Black condition ($M = 4.05$) had lower beliefs compared to participants in the Control ($M = 4.36$, $p = .001$), No Group ($M = 4.46$, $p < .001$), Working Class ($M = 4.31$, $p < .012$), and Poor conditions ($M = 4.39$, $p = .001$). No other comparisons were significant.

Sidak post hoc tests for support for communities impacted by heat waves revealed that participants in the Black condition ($M = 3.85$) had lower support compared to participants in the Poor condition ($M = 4.09$, $p = .048$). There were no significant differences in comparisons with participants in the Control ($M = 4.01$), No Group ($M = 4.06$), or Working Class ($M = 3.93$) conditions, and no other comparisons were significant.

Sidak post hoc tests for support for communities impacted by climate change revealed that participants in the Black condition ($M = 3.77$) had lower support compared to participants in the No Group ($M = 4.04$, $p = .031$) and Poor conditions ($M = 4.04$, $p = .035$). There were no

significant differences in comparisons with participants in the Control ($M$ = 3.95) or Working Class ($M$ = 3.83) conditions, and no other comparisons were significant.

***Symbolic Racism***

As with Study 1, we used hierarchical regression to analyze interactions between symbolic racism and the conditions. We first entered a block that included all main effects and then entered a second block that included the interaction between dummy coded message conditions and symbolic racism.

Looking first to main effects (*H1*), symbolic racism had a significant negative association with all dependent variables: belief in unequal impacts of heat waves (unstandardized $b$ = -.217, $p$ < .001), belief in unequal impacts of climate change ($b$ = -.358 , $p$ < .001), support for communities impacted by heat waves ($b$ = -.417 $p$ < .001), support for communities impacted by climate change ($b$ = -.494, $p$ < .001), support for policies to address heat waves ($b$ = -.350, $p$ < .001), and support for policies to address climate change ($b$ = -.467, $p$ < .001). The OLS regression table for symbolic racism can be found in Supplemental Material Table S7.

Including the interaction terms (*H*2a and *H*2b) only created a significant $R^2$ change for beliefs in unequal impacts of heat waves ($\Delta R^2$ = .022, $p$ < .001). Among individuals with high levels of symbolic racism, exposure to the Black condition led to lower levels of belief than all other conditions; for individuals with high levels of symbolic racism, the Poor condition also led to lower beliefs than the No Group condition. For individuals with low levels of symbolic racism, the Poor condition led to higher beliefs than the Control condition. No other interactions were significant. See Table 3 for regression coefficients and Johnson-Neyman results; see Figure 5 for a visual representation of all interactions. The full regression tables for all interactions

between conditions and symbolic racism can be found in Supplemental Material Tables S8 – S11.

***Race***

Next, we used ANOVAs to analyze the associations of race with the dependent variables (*H3*) as well as how race may moderate message effects (*H*4a and *H*4b). As with Study 1, for the race comparisons we only included participants ($n$ = 1,290) who identified as Black only or White only, as the sample size for other races was not sufficient for meaningful analysis.

The ANOVA revealed that race had a significant association with five of six dependent variables, such that Whites had lower beliefs, support for impacted communities and policy support than Blacks: beliefs in unequal impacts of climate change (F[1,1280] = 7.47, $p$ = .006, $\eta_p^2$ = .006; $M_{White}$ = 3.92, $M_{Black}$ = 4.10), support for communities impacted by heat waves (F[1, 1280] = 74.12, $p$ < .001, $\eta_p^2$ = .055; $M_{White}$ = 3.82, $M_{Black}$ = 4.30), support for communities impacted by climate change (F[1,1278] = 75.34, $p$ < .001, $\eta_p^2$ = .056; $M_{White}$ = 3.74, $M_{Black}$ = 4.27), support for policies to address heat waves (F[1, 1279] = 39.23, $p$ < .001, $\eta_p^2$ = .030; $M_{White}$ = 4.05, $M_{Black}$ = 4.37), and support for policies to address climate change (F[1,1278] = 30.99, $p$ < .001, $\eta_p^2$ = .024; $M_{White}$ = 3.82, $M_{Black}$ = 4.16).

The analysis failed to detect any significant interactions between message condition and race.

## Discussion

To the best of our knowledge, Studies 1 and 2 provide the first investigation into how exposure to news stories about unequal impacts of climate change due to race or class may influence public perceptions, and how message effects may be moderated by race and symbolic racism.

In Study 1, which focused on flooding risks, beliefs about unequal climate and flooding impacts and support for communities impacted by climate change were sensitive to whether race or class were emphasized in news stories (*RQ1)*. Where significant effects occurred, the general pattern was that beliefs and support for action were lower when the news story emphasized unequal impacts on Black communities. We also note that participants in the Poor condition had higher beliefs in unequal impacts of climate change and support for communities impacted by climate change than the Control condition, and that participants in the Working Class condition had higher beliefs in unequal impacts of climate change but lower beliefs in unequal impacts of flooding than participants in the Control condition.

Symbolic racism had a negative association with all dependent variables (in support of *H1*) and moderated the effects of the experimental condition for five of the six variables in Study 1 (in partial support of *H*2a and *H*2b). The general pattern was that the Black condition had negative effects for individuals higher in symbolic racism and some positive effects for those lower in symbolic racism, but the negative effects occurred more frequently.

In Study 1, participant race had a significant relationship with five of the six dependent variables (in partial support of *H3*), such that Black participants had stronger beliefs and support for action than Whites. Race moderated message effects for four of the dependent variables (in partial support of *H*4a and *H*4b), such that where moderation occurred there was no difference between conditions for Blacks, whereas White participants had lower beliefs in unequal impacts and support for impacted communities in the Black condition compared to several other conditions. White participants also had more support for communities impacted by climate change in the Poor and Working Class conditions than in the Control condition and also had

more support for policies to address flooding in the Working Class condition than the Control condition.

Similar to Study 1, in Study 2, which focused on heat waves, there were main effects of the experimental conditions on three of the dependent variables (*RQ1*), such that participants in the Black condition generally had lower beliefs and support for action. However, in Study 2, the moderating effects of symbolic racism and race were less consistent than in Study 1. In Study 2, as with Study 1, symbolic racism had a significant negative association with all dependent variables (in support of *H1*), but only moderated message effects on the belief in unequal impacts of heat waves (in partial support of *H*2a and *H*2b). Black participants generally had higher beliefs and support for action compared to White participants (in support of *H3*), but unlike Study 1, in Study 2 the analysis did not detect interactions between condition and race for any dependent variable (failing to support *H*4a and *H*4b).

Overall, this study makes several important contributions to the literature. First, the results suggest that messages about the unequal impacts of climate change may be less effective in increasing knowledge and support for action when they focus on racial disparities versus class-based disparities. These findings are consistent with the health disparities literature (Liu & Niederdeppe, 2024) and offer a valuable extension to the climate change domain. Although more research is needed to understand the precise mechanisms for these differences, our findings align with previous literature suggesting that information about racial disparities could increase stigmatization (Pearson et al., 2024) and stereotyping, leading some individuals to blame vulnerable groups for the climate impacts they experience (Swim & Bloodhart, 2018) and to minimize the harms these groups face.

The results also indicate that symbolic racism and race have significant associations with beliefs in unequal impacts and support for action to address these inequities. Consistent with previous literature, both studies found that symbolic racism is associated with lower levels of belief and support for action (Benegal, 2018; Chanin, 2018; Dietz et al., 2018) and that Blacks are likely to have higher levels of beliefs in unequal impacts and support for action than Whites (Pearson et al., 2018; Pearson & Schuldt, 2015). These findings point to the social factors that underpin beliefs about climate justice.

Consistent with social identity theories (Hornsey, 2008), the results also demonstrated that symbolic racism and the race of message recipients can moderate receptivity to narratives about unequal climate impacts. Whites and those higher in symbolic racism were more likely to react negatively to messages that highlight the vulnerabilities of Black communities, perhaps because these messages activated ethnocentrism and out-group derogation (Shanaah et al., 2024; Uhl et al., 2018). These findings echo past health disparities research showing that narratives emphasizing race-based vulnerabilities may backfire among certain audiences (Harell & Lieberman, 2021). Information about class-based vulnerabilities was less likely to activate these biases. In contrast to our results, one previous study, conducted in the UK, did find polarizing effects of a class-based climate justice narrative (Whitmarsh & Corner, 2017). However, their stimulus focused on climate harms faced by the global poor and attributed responsibility to wealthier countries and individuals—this attribution of responsibility may have alienated some audiences. A productive avenue for future research would be to explore how attribution of responsibility affects audience responses to climate inequality narratives.

It is also important to note that moderation by symbolic racism was more pronounced in Study 1 and race did not moderate effects in Study 2. Information about the vulnerabilities of

Black communities only infrequently boosted the beliefs and support of those low in symbolic racism, and did not alter beliefs and support for Black participants, who based on social identity processes may have been more receptive to such messaging. To the degree that social identity moderates message receptiveness, it appears more likely to have a negative rather than a positive relationship in the context of unequal climate impacts. It is important to note, however, that while White participants had lower mean scores than Black participants, and this was especially true when exposed to the Black condition in Study 1, White beliefs and support for action were still moderately positive – White participant mean scores were above 3.6 (on a 5-point scale) across all conditions and dependent variables in Study 1 and Study 2.

Notably, as mentioned above, the moderation results were not consistent across studies. Whereas symbolic racism and race moderated message effects for multiple dependent variables in Study 1, in Study 2 symbolic racism only had a significant interaction with messages on beliefs in unequal impacts and the race of the participant did not have any moderating effects. These discrepant results emphasize the importance of studying climate communication effects in different risk contexts and illuminate the contextual nature of social identity processes. It is possible that because flooding is seen as a more localized risk (Lohmann & Kontoleon, 2023), individuals who do not live in flood prone areas may feel less susceptible to flooding risks and less connected to those who are more at risk. This, in turn, may allow individuals with higher levels of symbolic racism and who do not share racial characteristics with those affected to more easily view Black communities impacted by climate change as an outgroup. In contrast, personal experience with heat waves, which is a risk that is less likely to be seen as localized (Lohmann & Kontoleon, 2023), may cause individuals to feel like they have more in common with those impacted by heat waves. Future research may benefit by exploring how specific characteristics of

risks such as flooding and heat waves, and personal experiences with these risks, influence the likelihood of race and symbolic racism moderating the effects of inequality messages.

For all observed effects, the effect sizes tended to be small. This is not surprising, as previous literature has found that across multiple types of stimuli, effect sizes for a single exposure to climate change messages tend to be very small (Rode et al., 2021). It is important to note that results may have differed if a stronger connection was drawn between experimental participants and impacted communities – for example, if the stimulus materials focused on a specific region from where experimental participants were also recruited (Hart & Nisbet, 2012). Future research should also consider the effects of repeated exposure to messages about unequal impacts of climate change, as well as over-time effects.

Strengths of this study include the use of carefully controlled experimental conditions, a test of the conditions across two specific types of climate risks, and a diverse national sample. However, as with any study, there are important limitations to consider. First, while the stimuli focused on two types of impacts to improve generalizability, it is still possible that specific wording within the news articles may have influenced the results and that news stories focusing on similar impacts, but with different wording, may generate different results. Second, all participants were based in the U.S., and the findings here may not generalize well to other countries. Third, the dependent variables used in both studies examined beliefs and support for action, but do not measure actual behavior (e.g. actual voting behavior or political action); the results observed here cannot directly speak to how exposure to similar news stories may influence actual behavior. Fourth, as noted in the Methodology section, for interactions the sample sizes were sufficient to detect medium effects, but the studies were not sufficiently powered to detect small effects; it is possible that small interaction effects occurred but were not

detected due to power limitations. Finally, while message effects were based on an experimental design that demonstrates causality, the associations of race and symbolic racism with the dependent variables are based on cross-sectional data, for which it is not possible to firmly establish a causal relationship.

Overall, the findings in Study 1 and Study 2 demonstrate that public beliefs about unequal climate impacts and support for impacted communities and policy support are sensitive to whether class or race is emphasized in climate change messaging, and that some of these effects are moderated by race and symbolic racism, although such moderation was more consistent in Study 1 than Study 2. As the impacts of climate change continue to grow, exacerbating vulnerabilities among marginalized communities, public support and political action for climate justice initiatives are critical (Carman et al., 2025). To this end, it is vital that news organizations accurately cover the disproportionate climate impacts faced by poor communities and communities of color. However, our results suggest that due to social identity processes, some Americans may not be receptive to this information, particularly when it emphasizes vulnerabilities associated with race rather than class. On the one hand, these results suggest that focusing on class-based climate disparities may be a more effective way to communicate about climate justice. On the other hand, journalists and other communicators must not ignore the unequal climate harms faced by communities of color, as these are real and must be mitigated. Thus, an urgent question for future research is how to develop communication strategies that can circumvent social identity processes and minimize biases to help advance climate justice and reduce the unequal harms of climate change.

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**Table 1**

*Measurement Text and Descriptives for Study1 and Study 2*

| | Items | Scale | Descriptives (Study 1) | Descriptives (Study 2) |
|---|---|---|---|---|
| Beliefs in unequal flooding (Study 1) / heat wave (Study 2) impacts | *Flooding / Heat waves affect(s) some communities more than other communities | Disagree strongly (1) ~ Agree strongly (5) | *M* = 4.47<br>*SD* = 0.82 | *M* = 4.31<br>*SD* = 0.97 |
| Beliefs in unequal climate change impacts | *Climate change affects some communities more than other communities. | Disagree strongly (1) ~ Agree strongly (5) | *M* = 3.86<br>*SD* = 1.26 | *M = 4.00*<br>*SD* = 1.19 |
| Support for communities impacted by flooding (Study 1) / heat wave (Study 2) | *How deserving of government aid are communities that are being impacted by flooding / heat waves?<br>*How important is it to help communities adapt to the impacts of flooding / heat waves?<br>*How much do you support or oppose increasing government spending to help communities adapt to the impacts of flooding / heat waves? | *Not at all deserving (1) ~ Extremely Deserving (5)<br>*Not at all important (1) ~ Extremely Important (5)<br>*Oppose strongly (1) ~ Support Strongly (5) | *M* = 4.22<br>*SD* = 0.80<br>Cronbach's $\alpha$ = .80 | *M* = 3.99<br>*SD* = 1.01<br>Cronbach's $\alpha$ = .87 |
| Support for communities impacted by climate change | *How deserving of government aid are communities that are being impacted by climate change?<br>*How important is it to help communities adapt to the impacts of climate change?<br>*How much do you support or oppose increasing government spending to help communities adapt to the impacts of climate change? | *Not at all deserving (1) ~ Extremely Deserving (5)<br>*Not at all important (1) ~ Extremely Important (5)<br>*Oppose strongly (1) ~ Support Strongly (5) | *M* = 3.96<br>*SD* = 1.10<br>Cronbach's $\alpha$ = .93 | *M* = 3.93<br>*SD* = 1.11<br>Cronbach's $\alpha$ = .92 |

| | | | | |
|---|---|---|---|---|
| Support for flooding policies (Study 1 only) | *Strengthen flood barriers, such as levees, in flood prone areas<br>*Widen natural flood plains in flood prone areas<br>*Use federal funds to protect and expand wetlands in flood prone areas<br>*Provide federal funds for the government to proactively offer to buy and remove homes in flood prone areas | Oppose strongly (1) ~ Support Strongly (5) | $M$ = 4.25<br>$SD$ = 0.73<br>Cronbach's $\alpha$ = .78 | *N/A* |
| Support for heat wave policies (Study 2 only) | *Provide more cooling centers in cities and towns when heat waves occur<br>*Increase government funding to plant more trees and vegetation in public spacing<br>*Use government funding to help individuals install green roofs on their homes<br>*Use government funding to build green infrastructure improvements, such as structures that can provide more shade | Oppose strongly (1) ~ Support Strongly (5) | *N/A* | $M$ = 4.16<br>$SD$ = 0.92<br>Cronbach's $\alpha$ = .87 |
| Support for climate change policies | *Regulate carbon dioxide and other greenhouse gases as pollutants<br>*Regulate greenhouse gas emissions from power plants<br>*Increase government investment in renewable energy industries like wind and solar<br>*Increase automobile fuel economy standards<br>*Redirect fuel subsidies to renewable energy | Oppose strongly (1) ~ Support Strongly (5) | $M$ = 3.98<br>$SD$ = 1.06<br>Cronbach's $\alpha$ = .93 | $M$ = 3.94<br>$SD$ = 1.09<br>Cronbach's $\alpha$ = .94 |
| Symbolic racism | *Many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors.<br>*It's really a matter of some people not trying hard enough; if Blacks would only try harder | Disagree strongly (1) ~ Agree strongly (5) | $M$ = 2.69<br>$SD$ = 1.21<br>Cronbach's $\alpha$ = .87 | $M$ = 2.70<br>$SD$ = 1.24<br>Cronbach's $\alpha$ = .88 |

| | | | | |
|---|---|---|---|---|
| | they could be just as well off as Whites.<br>*Generations of slavery and discrimination have created conditions that make it difficult for Blacks to work their way out of the lower class. (R)<br>*Over the past few years, Blacks have gotten less than they deserve. (R) | | | |

*Note.* Questions for beliefs in unequal impacts were adapted from Schuldt and Pearson (2023). Questions for support for climate change policies were adapted from Hall et al. (2018) and Feldman and Hart (2018, 2021). Questions for symbolic racism were adapted from the four-item symbolic racism scale in ANES 2020 (American National Election Studies, 2021).

**Table 2**

*ANOVA Results and Means Comparisons Across Conditions for Study 1*

| | Condition | | | | | Main Effect *F* from one-way ANOVA |
|---|---|---|---|---|---|---|
| | Control *M* (*SD*) | No Group *M* (*SD*) | Black *M* (*SD*) | Poor *M* (*SD*) | Working *M* (*SD*) | |
| Dependent Variable | | | | | | |
| Belief in unequal impacts - flooding | 4.65 (0.67)$_a$ | 4.51 (0.73)$_{a,c}$ | 4.23 (0.99)$_b$ | 4.51 (0.77)$_{a,c}$ | 4.43 (0.87)$_c$ | 10.36, $p < .001$; $\eta^2 = .03$ |
| Belief in unequal impacts - climate change | 3.67 (1.36)$_a$ | 3.85 (1.23) | 3.74 (1.32)$_{a,c}$ | 4.06 (1.20)$_b$ | 3.99 (1.19)$_{b,c}$ | 4.88, $p < .001$; $\eta^2 = .01$ |
| Comm. support - flooding | 4.22 (0.76) | 4.19 (0.86) | 4.13 (0.88)$_a$ | 4.32 (0.74)$_b$ | 4.25 (0.76) | 2.19, $p = .068$; $\eta^2 = .006$ |
| Comm. support - climate | 3.79 (1.13)$_a$ | 3.80 (1.09) | 3.92 (1.18) | 4.07 (1.04)$_b$ | 4.02 (1.02) | 2.83, $p = .023$; $\eta^2 = .008$ |
| Flooding policy support | 4.17 (0.72) | 4.22 (0.79) | 4.28 (0.73) | 4.31 (0.65) | 4.28 (0.74) | 1.80, $p = .127$; $\eta^2 = .005$ |
| Climate policy support | 3.91 (1.06) | 3.93 (1.11) | 3.98 (1.06) | 4.02 (1.02) | 4.04 (1.03) | 0.84, $p = .500$; $\eta^2 = .002$ |
| *N* | 294 | 286 | 281 | 280 | 293 | 1,434 |

*Note.* Within rows, means with different letter subscripts are significantly different from one another, $p < .05$, after the Sidak adjustment. Means without a letters subscript are not significantly different from any other condition.

**Table 3**

*Unstandardized coefficients and p-value for the interaction between symbolic racism and message condition, as well as Johnson-Neyman Results, for Study 1 and Study 2.*

| | Study 1 | | | | | | Study 2 | |
|---|---|---|---|---|---|---|---|---|
| | Belief in Unequal flooding impacts | Belief in Unequal climate impacts | Support for aid to communities impacted by flooding | Support for aid to communities impacted by climate change | Support for flooding policies | Support for climate policies | Belief in Unequal heat wave impacts | All other DVs |
| Control vs. Black | *b* = .312, *p* < .001<br>JN: >1.94 | ns | *b* = .209, *p* < .001<br>JN: <1.68, >3.02 | ns | *b* = .154, *p* = .001<br>JN: <2.92 | ns | *b* = .290, *p* < .001<br>JN: >2.38 | ns |
| No Group vs. Black | *b* = .228, *p* < .001<br>JN: >2.23 | *b* = .432, *p* < .002<br>JN: <2.02, >3.03 | ns | *b* = .183, *p* = .005<br>JN: >3.33 | ns | ns | *b* = .311, *p* < .001<br>JN: >2.14 | ns |
| Poor vs. Black | *b* = .199, *p* < .001<br>JN: >2.15 | ns | *b* = .135, *p* = .008<br>>2.34 | *b* = .166, *p* = .011<br>>2.82 | ns | ns | *b* = .156, *p* = .012<br>JN: >1.96 | ns |
| Working vs. Black | *b* = .182, *p* < .001<br>JN: >2.64 | *b* = .249, *p* = .002<br>JN: >2.90 | *b* = .138, *p* = .006<br>JN: >3.28 | *b* = .187, *p* = .004<br>JN: <1.70, >3.96 | *b* = .129, *p* = .007<br>JN: <2.17 | ns | *b* = .240, *p* < .001<br>JN: >2.43 | ns |
| Control vs. Poor | *b* = .112, *p* = .036<br>JN: >2.68 | ns | ns | ns | ns | ns | *b* = .133, *p* = .028<br>JN: <1.56 | ns |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| No Group vs. Poor | ns | *b* = .280, *p* < .001<br>JN: <2.80 | ns | ns | ns | ns | *b* = .155, *p* = .016<br>JN: >3.73 | ns |
| Working vs. Poor | ns | Ns | ns | ns | ns | ns | ns | ns |
| Control vs. Working | *b* = .130, *p* = .015<br>JN: >1.92 | ns | ns | ns | ns | ns | ns | ns |
| No Group vs. Working | ns | *b* = .182, *p* = .029<br>JN: <1.96 | ns | ns | ns | ns | ns | ns |
| Control vs. No Group | ns | *b* = -.288, *p* < .001<br>JN: >2.67 | *b* =.119, *p* = .020<br>JN: >4.42 | ns | *b* =.094, *p* = .050<br>JN: <2.00 | ns | ns | ns |

*Note.* The *top row* of cell entries represent the unstandardized coefficients and p-value for the interaction between symbolic racism and message condition comparison. *The bottom row* represent the range of values on the symbolic racism scale for which the comparison between conditions is significant, based on a Johnson-Neyman analysis. Shaded values represent a positive difference between conditions, meaning that the first condition had a higher value than the second condition; unshaded values represent a negative difference between conditions. ns = interaction was not significant

**Table 4**

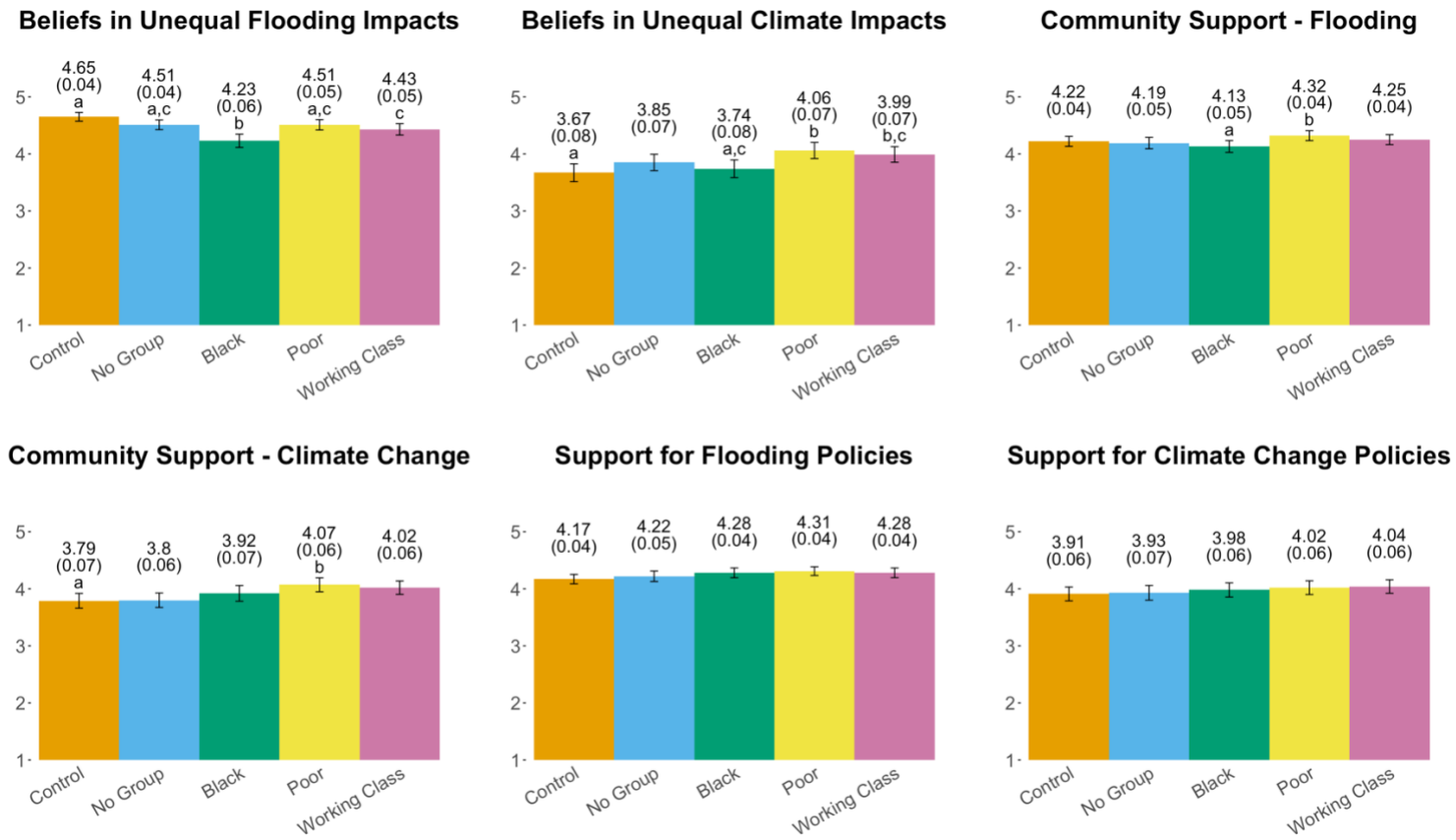
*ANOVA Results and Means Comparisons Across Conditions for Study 2*

| | Condition | | | | | Main Effect *F* from one-way ANOVA |
|---|---|---|---|---|---|---|
| | Control *M* (*SD*) | No Group *M* (*SD*) | Black *M* (*SD*) | Poor *M* (*SD*) | Working *M* (*SD*) | |
| Dependent Variable | | | | | | |
| Belief in unequal impacts – heat waves | 4.36 (0.88)$_a$ | 4.46 (0.85)$_a$ | 4.05 (1.17)$_b$ | 4.39 (0.93)$_a$ | 4.31 (0.92)$_a$ | 7.69, $p < .001$; $\eta^2 = .02$ |
| Belief in unequal impacts – climate change | 3.96 (1.21) | 4.06 (1.18) | 3.91 (1.25) | 4.08 (1.14) | 3.97 (1.17) | 1.04, $p = .383$; $\eta^2 = .003$ |
| Comm. support – heat waves | 4.01 (0.98) | 4.06 (0.94) | 3.85 (1.11)$_a$ | 4.09 (0.99)$_b$ | 3.93 (0.99) | 2.64, $p = .032$; $\eta^2 = .007$ |
| Comm. support - climate | 3.95 (1.11) | 4.04 (1.06)$_a$ | 3.77 (1.19)$_b$ | 4.04 (1.09)$_a$ | 3.84 (1.09) | 3.45, $p = .008$; $\eta^2 = .01$ |
| Heat wave policy support | 4.18 (0.86) | 4.23 (0.87) | 4.08 (0.99) | 4.18 (0.94) | 4.14 (0.92) | 1.10, $p = .355$; $\eta^2 = .003$ |
| Climate policy support | 3.98 (1.06) | 4.05 (1.10) | 3.83 (1.11) | 3.97 (1.10) | 3.86 (1.08) | 2.14, $p = .074$; $\eta^2 = .006$ |
| *N* | 298 | 282 | 285 | 276 | 280 | 1,421 |

*Note.* Within rows, means with different letter subscripts are significantly different from one another, $p < .05$, after the Sidak adjustment. Means without a letters subscript are not significantly different from any other condition.

**Figure 1**

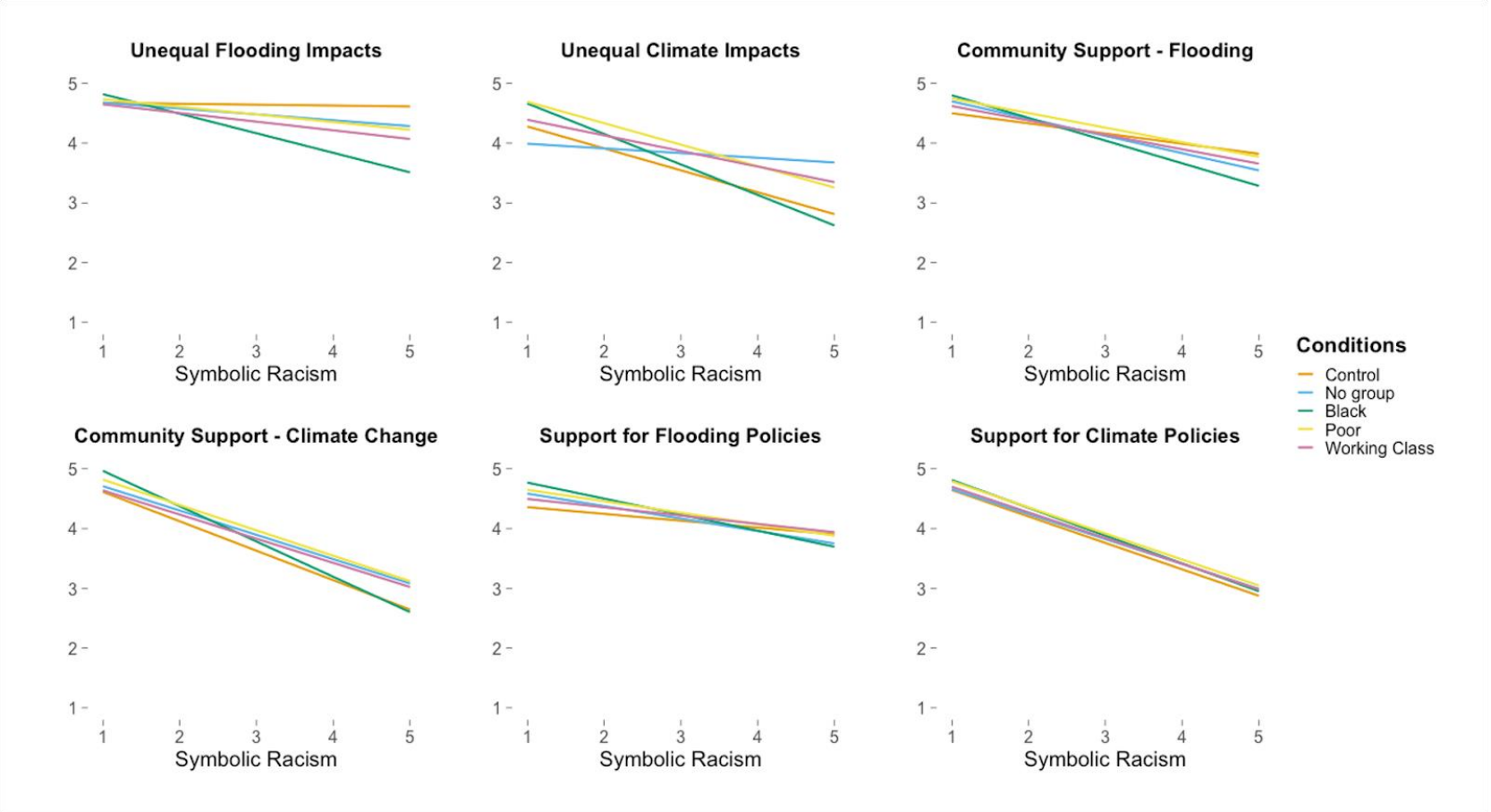
*Mean Scores by Condition for Study 1*




*Note.* Mean scores by condition for the six dependent variables are shown. Above each column, the top number is the mean score, the number in parentheses is the standard error, and the bars above and below the means are the 95% confidence intervals. Means with different letter subscripts are significantly different from one another, $p < .05$, after the Sidak adjustment. Means without a letters subscript are not significantly different from any other condition.

**Figure 2**

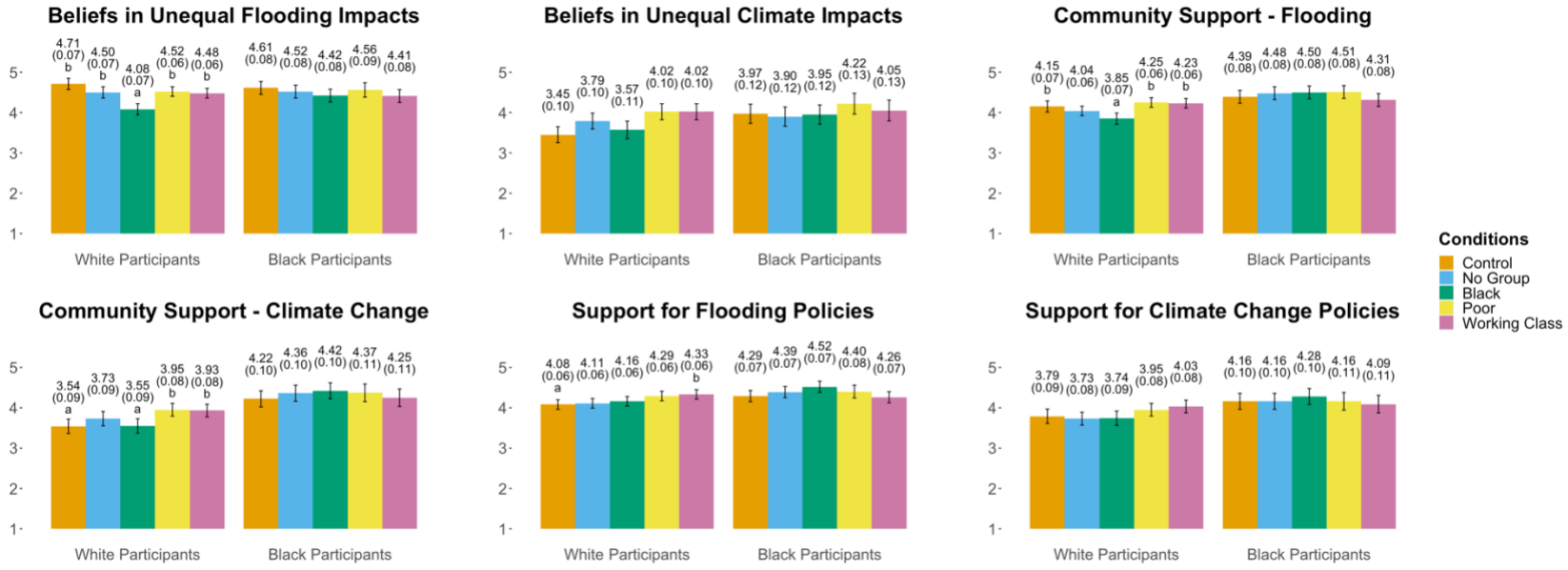
*Interactions Between Symbolic Racism and Condition for Study 1*




*Note.* Including the interaction terms created a significant $R^2$ change, $p < .05$, for five of the six dependent variables: belief in unequal impacts of flooding, belief in unequal impacts of climate change, support for communities impacted by flooding, support for communities impacted by climate change, and support for policies to address flooding.

**Figure 3**

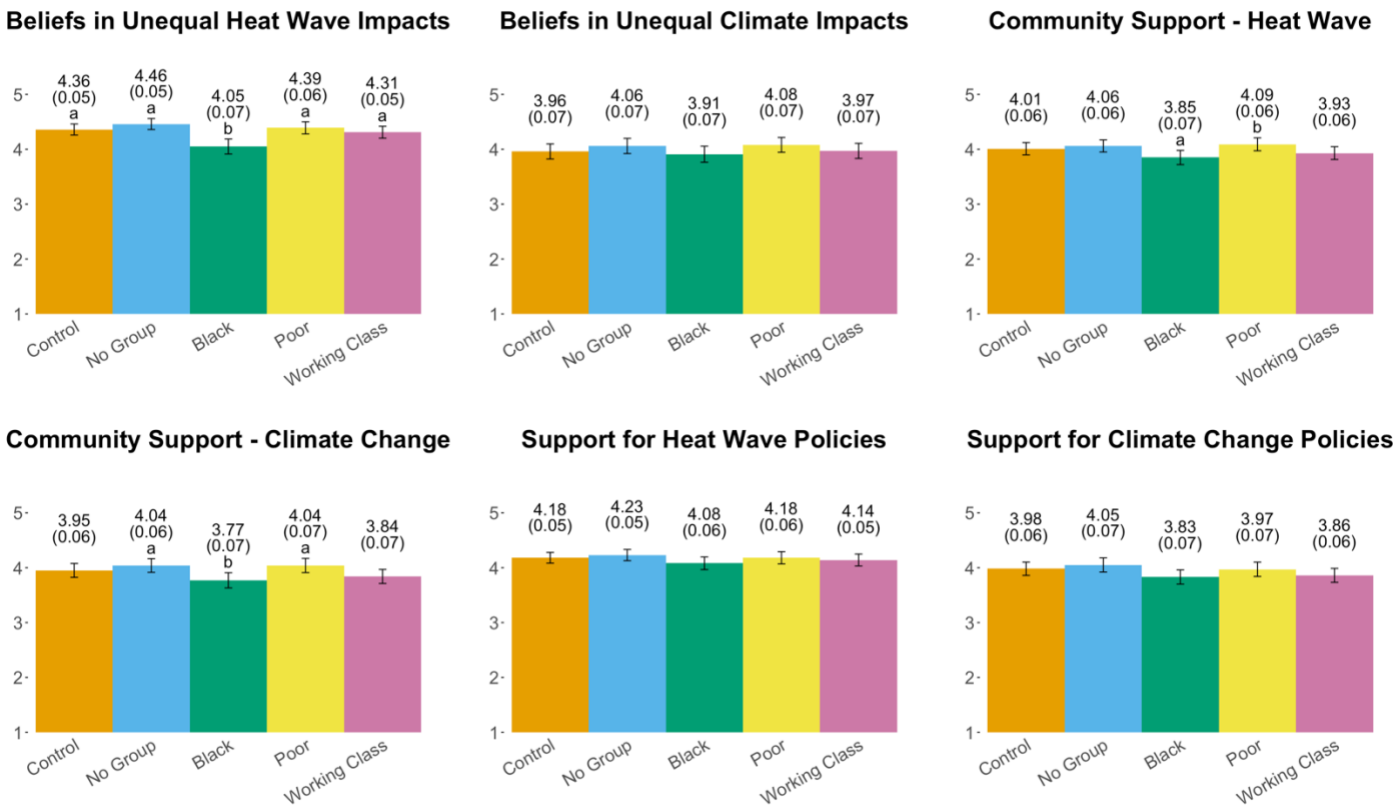
*Interaction Between Race and Condition for Study 1*




*Note.* Mean scores by condition and race for the six dependent variables are shown. Above each column, the top number is the mean score, the number in parentheses is the standard error, and the bars above and below the means are the 95% confidence intervals. Means with different letter subscripts are significantly different from one another, $p < .05$, after the Sidak adjustment. Means without a letter subscript are not significantly different from any other condition.

**Figure 4**

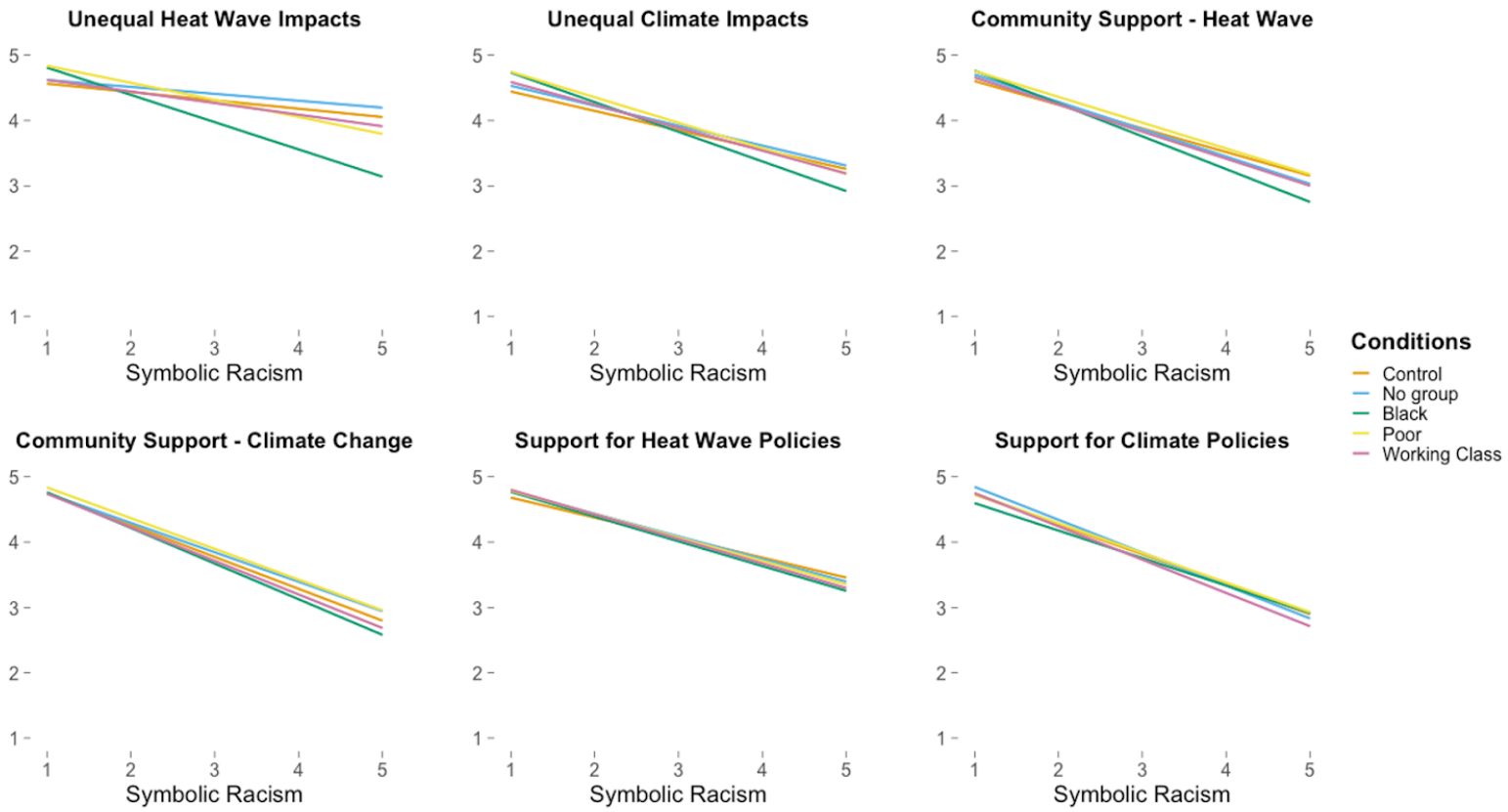
*Mean Scores by Condition for Study 2*




*Note.* Mean scores by condition for the six dependent variables are shown. Above each column, the top number is the mean score, the number in parentheses is the standard error, and the bars above and below the means are the 95% confidence intervals. Means with different letter subscripts are significantly different from one another, $p < .05$, after the Sidak adjustment. Means without a letter subscript are not significantly different from any other condition.

**Figure 5**

*Interactions Between Symbolic Racism and Condition for Study 2*




*Note.* Including the interaction terms only created a significant $R^2$ change, $p < .05$, for beliefs in unequal impacts of heat waves.

**Supplemental Material**

**Text Used in Conditions for Study 1 (Flooding) and Study 2 (Heat Waves)**

***Flooding: No Group Condition***



August 7, 2023 10:30 AM EDT

## New Study: More dangerous floods are on the way

BY FRANK REYNOLDS
ASSOCIATED PRESS



WASHINGTON (AP) - Flood risk in the United States will increase by about 25% in the next three decades, according to a sweeping new analysis published in Nature Climate Change. Climate change is already driving more severe flooding across much of the country. By 2050, annual losses from floods will be approximately $40 billion, according to the new study.

The study underscores the need to adapt to a hotter earth in addition to reducing greenhouse gas emissions that cause climate change. The authors of the new study stress that it is not too late to protect people from climate-driven flooding. They find that where people live is by far the most important factor for overall flood risk. Recently the federal government has launched a program to buy and remove homes in areas that have high levels of flood risk. Additional adaptive strategies include strengthening flood barriers, such as levees, as well as nature-based solutions such as widening natural flood plains and protecting and expanding wetlands.

***Flooding: Black / Poor / Working Class Conditions***

August 7, 2023 10:30 AM EDT

# New Study: More dangerous floods are on the way, with greater risk to (Black / poor / working class) communities

BY FRANK REYNOLDS
ASSOCIATED PRESS

WASHINGTON (AP) - Flood risk in the United States will increase by about 25% in the next three decades, and (Black / poor / working class) communities will face disproportionate harm, according to a sweeping new analysis published in Nature Climate Change.

The new study attempts to estimate not just the scale of flood risk in the U.S., but also who will bear the burden of flooding. The study finds that communities where (Black / poor / working class) people live will see a disproportionate rise in flood risk.

Climate change is already driving more severe flooding across much of the country. By 2050, annual losses from floods will be approximately $40 billion, according to the new study. The study finds that flood risks will be concentrated in (Black / poor / working class) neighborhoods that were designated as risky investments decades ago. As the global temperature rises, these (Black / poor / working class) communities will experience an increased risk of dangerous flooding.

The study underscores the need to adapt to a hotter earth in addition to reducing greenhouse gas emissions that cause climate change. The authors of the new study stress that it is not too late to protect people from climate-driven flooding. They find that where people live is by far the most important factor for overall flood risk. Recently the federal government has launched a program to buy and remove homes in areas that have high levels of flood risk. Additional adaptive strategies include strengthening flood barriers, such as levees, as well as nature-based solutions such as widening natural flood plains and protecting and expanding wetlands.

***Heat Waves: No Group Condition***



August 7, 2023 10:30 AM EDT

# New Study: More dangerous heat waves are on the way

BY FRANK REYNOLDS
ASSOCIATED PRESS



WASHINGTON (AP) - By mid-century, nearly two-thirds of Americans will experience perilous heat waves. Across much of the United States, millions of people are expected to experience extreme temperatures more frequently and for longer periods of time, according to a sweeping new analysis published in Nature Climate Change. Overall, the risk of heat waves is expected to increase dramatically due to climate change.

Heat is the top weather-related killer in the United States. Treeless city neighborhoods, packed with buildings, parking lots and asphalt roads, absorb and retain more heat than areas with tree-lined streets and parks. Scientists call this the urban heat island effect. Nationwide, this pattern reveals itself in city after city, concentrating heat.

The study underscores the need to adapt to a hotter earth in addition to reducing greenhouse gas emissions that cause climate change. The authors of the new study stress that it is not too late to protect people from climate-driven heat waves. To improve resilience to future extreme heat events, cities can incorporate heat island reduction strategies—such as green or cool roofs, cool pavements, or increased vegetation and trees—into long-term planning efforts to help lower urban temperatures. Such cooling measures help to reduce impacts on public health and urban systems from extreme heat events.

***Heat Waves: Black / Poor / Working Class Conditions***

August 7, 2023 10:30 AM EDT

# New Study: More dangerous heat waves are on the way, with greater risk to (Black / poor / working class) communities

BY FRANK REYNOLDS
ASSOCIATED PRESS

WASHINGTON (AP) - By mid-century, nearly two-thirds of Americans will experience perilous heat waves, with (Black / poor / working class) communities most at risk. Across much of the United States, millions of people are expected to experience extreme temperatures more frequently and for longer periods of time, according to a sweeping new analysis published in Nature Climate Change. The new study attempts to estimate not just the scale of heat wave risks in the U.S., but also who will bear the burden. The study finds that communities where (Black / poor / working class) people live will see a disproportionate rise in risks from heat waves. Overall, the risk of heat waves is expected to increase dramatically due to climate change.

Heat is the top weather-related killer in the United States. But like other effects of climate change, it is felt unevenly. (Black / poor / working class) communities are especially at risk. Treeless city neighborhoods, packed with buildings, parking lots and asphalt roads, absorb and retain more heat than areas with tree-lined streets and parks. Scientists call this the urban heat island effect. Nationwide, this pattern reveals itself in city after city, concentrating heat in (Black / poor / working class) neighborhoods that were designated as risky investments decades ago. As the global temperature rises, these (Black / poor / working class) communities will experience an increasing risk of dangerous heat waves.

The study underscores the need to adapt to a hotter earth in addition to reducing greenhouse gas emissions that cause climate change. The authors of the new study stress that it is not too late to protect people from climate-driven heat waves. To improve resilience to future extreme heat events, cities can incorporate heat island reduction strategies—such as green or cool roofs, cool pavements, or increased vegetation and trees—into long-term planning efforts to help lower urban temperatures. Such cooling measures help to reduce impacts on public health and urban systems from extreme heat events.

**Table S1**

*Study 1: Main Effects for Symbolic Racism and Stimulus Conditions Compared to the Control Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Flooding Impact | Unequal Climate Impact | Community Support - Flooding | Community Support - Climate | Policy Support - Flooding | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| No Group vs. Control | -0.131* | 0.211* | -0.002 | 0.241** | 0.066 | 0.058 |
| | (0.066) | (0.100) | (0.061) | (0.078) | (0.057) | (0.076) |
| Black vs. Control | -0.394*** | 0.126 | -0.061 | 0.174* | 0.144* | 0.127 |
| | (0.066) | (0.100) | (0.061) | (0.078) | (0.058) | (0.077) |
| Poor vs. Control | -0.126 | 0.423*** | 0.125* | 0.321*** | 0.159** | 0.157* |
| | (0.066) | (0.100) | (0.061) | (0.078) | (0.058) | (0.076) |
| Working vs. Control | -0.241*** | 0.282** | -0.001 | 0.164* | 0.089 | 0.075 |
| | (0.065) | (0.099) | (0.061) | (0.077) | (0.057) | (0.076) |
| Racism | -0.142*** | -0.318*** | -0.263*** | -0.464*** | -0.183*** | -0.438*** |
| | (0.017) | (0.026) | (0.016) | (0.021) | (0.015) | (0.020) |
| Constant | 5.029*** | 4.516*** | 4.919*** | 5.031*** | 4.657*** | 5.075*** |
| | (0.065) | (0.099) | (0.061) | (0.077) | (0.057) | (0.076) |
| Observations | 1,432 | 1,431 | 1,431 | 1,432 | 1,431 | 1,431 |
| $R^2$ | 0.071 | 0.105 | 0.163 | 0.269 | 0.098 | 0.252 |
| Adjusted $R^2$ | 0.067 | 0.102 | 0.160 | 0.267 | 0.094 | 0.250 |
| Residual Std. Error | 0.790 (df = 1426) | 1.201 (df = 1425) | 0.734 (df = 1425) | 0.938 (df = 1426) | 0.691 (df = 1425) | 0.916 (df = 1425) |
| F Statistic | 21.716*** (df = 5; 1426) | 33.420*** (df = 5; 1425) | 55.503*** (df = 5; 1425) | 105.171*** (df = 5; 1426) | 30.827*** (df = 5; 1425) | 96.084*** (df = 5; 1425) |

*Note*. The Control condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S2**

*Study 1: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Control Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Flooding Impact | Unequal Climate Impact | Community Support - Flooding | Community Support - Climate | Policy Support - Flooding | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| No Group vs. Control | 0.088 | -0.576* | 0.318* | 0.008 | 0.320* | -0.011 |
| | (0.161) | (0.246) | (0.151) | (0.193) | (0.142) | (0.189) |
| Black vs. Control | 0.456** | 0.528* | 0.510*** | 0.449* | 0.564*** | 0.190 |
| | (0.160) | (0.243) | (0.149) | (0.191) | (0.141) | (0.188) |
| Poor vs. Control | 0.173 | 0.406 | 0.320* | 0.137 | 0.368** | 0.140 |
| | (0.160) | (0.244) | (0.149) | (0.191) | (0.141) | (0.187) |
| Working vs. Control | 0.104 | 0.007 | 0.192 | -0.064 | 0.162 | 0.042 |
| | (0.153) | (0.233) | (0.143) | (0.183) | (0.135) | (0.179) |
| Racism | -0.015 | -0.367*** | -0.170*** | -0.492*** | -0.114*** | -0.442*** |
| | (0.037) | (0.057) | (0.035) | (0.045) | (0.033) | (0.044) |
| No Group * Racism | -0.083 | 0.288*** | -0.119* | 0.086 | -0.094* | 0.025 |
| | (0.055) | (0.083) | (0.051) | (0.065) | (0.048) | (0.064) |
| Black * Racism | -0.312*** | -0.143 | -0.209*** | -0.098 | -0.154** | -0.023 |
| | (0.053) | (0.082) | (0.050) | (0.064) | (0.047) | (0.063) |
| Poor * Racism | -0.112* | 0.008 | -0.074 | 0.068 | -0.078 | 0.006 |
| | (0.054) | (0.082) | (0.050) | (0.064) | (0.047) | (0.063) |
| Working * Racism | -0.130* | 0.106 | -0.071 | 0.088 | -0.025 | 0.013 |
| | (0.053) | (0.081) | (0.050) | (0.064) | (0.047) | (0.062) |
| Constant | 4.690*** | 4.645*** | 4.670*** | 5.106*** | 4.473*** | 5.086*** |
| | (0.110) | (0.168) | (0.102) | (0.131) | (0.097) | (0.129) |
| Observations | 1,432 | 1,431 | 1,431 | 1,432 | 1,431 | 1,431 |
| $R^2$ | 0.094 | 0.123 | 0.174 | 0.276 | 0.106 | 0.252 |
| Adjusted $R^2$ | 0.088 | 0.117 | 0.169 | 0.271 | 0.100 | 0.248 |
| Residual Std. Error | 0.782 (df = 1422) | 1.190 (df = 1421) | 0.730 (df = 1421) | 0.935 (df = 1422) | 0.689 (df = 1421) | 0.917 (df = 1421) |

| | | | | | | |
|---|---|---|---|---|---|---|
| F Statistic | 16.362*** (df = 9; 1422) | 22.080*** (df = 9; 1421) | 33.249*** (df = 9; 1421) | 60.091*** (df = 9; 1422) | 18.685*** (df = 9; 1421) | 53.320*** (df = 9; 1421) |

*Note*. The Control condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S3**

*Study 1: Interaction Between Symbolic Racism and Experimental Conditions Compared to the No Group Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Flooding Impact | Unequal Climate Impact | Community Support - Flooding | Community Support - Climate | Policy Support - Flooding | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. No Group | -0.088 | 0.576* | -0.318* | -0.008 | -0.320* | 0.011 |
| | (0.161) | (0.246) | (0.151) | (0.193) | (0.142) | (0.189) |
| Black vs. No Group | 0.368* | 1.104*** | 0.192 | 0.441* | 0.245 | 0.200 |
| | (0.166) | (0.252) | (0.155) | (0.198) | (0.146) | (0.195) |
| Poor vs. No Group | 0.085 | 0.982*** | 0.003 | 0.128 | 0.048 | 0.151 |
| | (0.166) | (0.253) | (0.155) | (0.198) | (0.146) | (0.195) |
| Working vs. No Group | 0.016 | 0.583* | -0.126 | -0.072 | -0.158 | 0.052 |
| | (0.159) | (0.242) | (0.149) | (0.190) | (0.140) | (0.187) |
| Racism | -0.099* | -0.078 | -0.289*** | -0.407*** | -0.208*** | -0.417*** |
| | (0.040) | (0.060) | (0.037) | (0.048) | (0.035) | (0.047) |
| Control * Racism | 0.083 | -0.288*** | 0.119* | -0.086 | 0.094* | -0.025 |
| | (0.055) | (0.083) | (0.051) | (0.065) | (0.048) | (0.064) |
| Black * Racism | -0.228*** | -0.431*** | -0.090 | -0.184** | -0.060 | -0.048 |
| | (0.055) | (0.084) | (0.052) | (0.066) | (0.049) | (0.065) |
| Poor * Racism | -0.029 | -0.280*** | 0.045 | -0.018 | 0.016 | -0.019 |
| | (0.055) | (0.084) | (0.052) | (0.066) | (0.049) | (0.065) |
| Working * Racism | -0.046 | -0.182* | 0.048 | 0.003 | 0.069 | -0.012 |
| | (0.055) | (0.083) | (0.051) | (0.066) | (0.048) | (0.064) |
| Constant | 4.778*** | 4.069*** | 4.988*** | 5.114*** | 4.793*** | 5.076*** |
| | (0.118) | (0.180) | (0.111) | (0.142) | (0.104) | (0.139) |
| Observations | 1,432 | 1,431 | 1,431 | 1,432 | 1,431 | 1,431 |
| $R^2$ | 0.094 | 0.123 | 0.174 | 0.276 | 0.106 | 0.252 |
| Adjusted $R^2$ | 0.088 | 0.117 | 0.169 | 0.271 | 0.100 | 0.248 |
| Residual Std. Error | 0.782 (df = 1422) | 1.190 (df = 1421) | 0.730 (df = 1421) | 0.935 (df = 1422) | 0.689 (df = 1421) | 0.917 (df = 1421) |
| F Statistic | 16.362*** (df = 9; 1422) | 22.080*** (df = 9; 1421) | 33.249*** (df = 9; 1421) | 60.091*** (df = 9; 1422) | 18.685*** (df = 9; 1421) | 53.320*** (df = 9; 1421) |

*Note*. The No Group condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S4**

*Study 1: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Black Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Flooding Impact | Unequal Climate Impact | Community Support - Flooding | Community Support - Climate | Policy Support - Flooding | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. Black | -0.456** | -0.528* | -0.510*** | -0.449* | -0.564*** | -0.190 |
| | (0.160) | (0.243) | (0.149) | (0.191) | (0.141) | (0.188) |
| No Group vs. Black | -0.368* | -1.104*** | -0.192 | -0.441* | -0.245 | -0.200 |
| | (0.166) | (0.252) | (0.155) | (0.198) | (0.146) | (0.195) |
| Poor vs. Black | -0.283 | -0.122 | -0.189 | -0.312 | -0.196 | -0.050 |
| | (0.164) | (0.250) | (0.153) | (0.196) | (0.145) | (0.193) |
| Working Class vs. Black | -0.352* | -0.521* | -0.318* | -0.513** | -0.403** | -0.148 |
| | (0.157) | (0.239) | (0.147) | (0.188) | (0.139) | (0.185) |
| Racism | -0.327*** | -0.510*** | -0.379*** | -0.590*** | -0.268*** | -0.465*** |
| | (0.038) | (0.058) | (0.036) | (0.046) | (0.034) | (0.045) |
| Control * Racism | 0.312*** | 0.143 | 0.209*** | 0.098 | 0.154** | 0.023 |
| | (0.053) | (0.082) | (0.050) | (0.064) | (0.047) | (0.063) |
| No Group * Racism | 0.228*** | 0.431*** | 0.090 | 0.184** | 0.060 | 0.048 |
| | (0.055) | (0.084) | (0.052) | (0.066) | (0.049) | (0.065) |
| Poor * Racism | 0.199*** | 0.151 | 0.135** | 0.166* | 0.076 | 0.029 |
| | (0.054) | (0.083) | (0.051) | (0.065) | (0.048) | (0.064) |
| Working * Racism | 0.182*** | 0.249** | 0.138** | 0.187** | 0.129** | 0.036 |
| | (0.054) | (0.082) | (0.050) | (0.064) | (0.047) | (0.063) |
| Constant | 5.146*** | 5.173*** | 5.180*** | 5.555*** | 5.037*** | 5.276*** |
| | (0.116) | (0.176) | (0.108) | (0.139) | (0.103) | (0.137) |
| Observations | 1,432 | 1,431 | 1,431 | 1,432 | 1,431 | 1,431 |
| $R^2$ | 0.094 | 0.123 | 0.174 | 0.276 | 0.106 | 0.252 |
| Adjusted $R^2$ | 0.088 | 0.117 | 0.169 | 0.271 | 0.100 | 0.248 |
| Residual Std. Error | 0.782 (df = 1422) | 1.190 (df = 1421) | 0.730 (df = 1421) | 0.935 (df = 1422) | 0.689 (df = 1421) | 0.917 (df = 1421) |
| F Statistic | 16.362*** (df = 9; 1422) | 22.080*** (df = 9; 1421) | 33.249*** (df = 9; 1421) | 60.091*** (df = 9; 1422) | 18.685*** (df = 9; 1421) | 53.320*** (df = 9; 1421) |

*Note*. The Black condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S5**

*Study 1: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Poor Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Flooding Impact | Unequal Climate Impact | Community Support - Flooding | Community Support - Climate | Policy Support - Flooding | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. Poor | -0.173 | -0.406 | -0.320* | -0.137 | -0.368** | -0.140 |
| | (0.160) | (0.244) | (0.149) | (0.191) | (0.141) | (0.187) |
| No Group vs. Poor | -0.085 | -0.982*** | -0.003 | -0.128 | -0.048 | -0.151 |
| | (0.166) | (0.253) | (0.155) | (0.198) | (0.146) | (0.195) |
| Black vs. Poor | 0.283 | 0.122 | 0.189 | 0.312 | 0.196 | 0.050 |
| | (0.164) | (0.250) | (0.153) | (0.196) | (0.145) | (0.193) |
| Working vs. Poor | -0.069 | -0.399 | -0.129 | -0.201 | -0.206 | -0.098 |
| | (0.157) | (0.239) | (0.147) | (0.188) | (0.139) | (0.184) |
| Racism | -0.128*** | -0.359*** | -0.244*** | -0.424*** | -0.192*** | -0.436*** |
| | (0.038) | (0.059) | (0.036) | (0.046) | (0.034) | (0.045) |
| Control * Racism | 0.112* | -0.008 | 0.074 | -0.068 | 0.078 | -0.006 |
| | (0.054) | (0.082) | (0.050) | (0.064) | (0.047) | (0.063) |
| No Group * Racism | 0.029 | 0.280*** | -0.045 | 0.018 | -0.016 | 0.019 |
| | (0.055) | (0.084) | (0.052) | (0.066) | (0.049) | (0.065) |
| Black * Racism | -0.199*** | -0.151 | -0.135** | -0.166* | -0.076 | -0.029 |
| | (0.054) | (0.083) | (0.051) | (0.065) | (0.048) | (0.064) |
| Working * Racism | -0.017 | 0.098 | 0.003 | 0.021 | 0.053 | 0.007 |
| | (0.054) | (0.082) | (0.050) | (0.064) | (0.048) | (0.063) |
| Constant | 4.863*** | 5.051*** | 4.990*** | 5.243*** | 4.841*** | 5.226*** |
| | (0.116) | (0.177) | (0.108) | (0.139) | (0.102) | (0.136) |
| Observations | 1,432 | 1,431 | 1,431 | 1,432 | 1,431 | 1,431 |
| $R^2$ | 0.094 | 0.123 | 0.174 | 0.276 | 0.106 | 0.252 |
| Adjusted $R^2$ | 0.088 | 0.117 | 0.169 | 0.271 | 0.100 | 0.248 |
| Residual Std. Error | 0.782 (df = 1422) | 1.190 (df = 1421) | 0.730 (df = 1421) | 0.935 (df = 1422) | 0.689 (df = 1421) | 0.917 (df = 1421) |

| | | | | | | |
|---|---|---|---|---|---|---|
| F Statistic | 16.362*** (df = 9; 1422) | 22.080*** (df = 9; 1421) | 33.249*** (df = 9; 1421) | 60.091*** (df = 9; 1422) | 18.685*** (df = 9; 1421) | 53.320*** (df = 9; 1421) |

*Note*. The Poor condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S6**

*Mean scores by Race and Condition for Study 1.*

| Dependent Variable | Participant Race | Condition | M (SD) |
|---|---|---|---|
| Belief in unequal impacts - flooding | White | Control | 4.71 (0.56) |
| | | No Group | 4.50 (0.79) |
| | | Black | 4.08 (1.14) |
| | | Poor | 4.52 (0.79) |
| | | Working | 4.48 (0.89) |
| | Black | Control | 4.61 (0.70) |
| | | No Group | 4.51 (0.64) |
| | | Black | 4.42 (0.80) |
| | | Poor | 4.56 (0.69) |
| | | Working | 4.41 (0.81) |
| Belief in unequal impacts – climate change | White | Control | 3.45 (1.44) |
| | | No Group | 3.79 (1.23) |
| | | Black | 3.57 (1.41) |
| | | Poor | 4.02 (1.25) |
| | | Working | 4.02 (1.22) |
| | Black | Control | 3.97 (1.17) |
| | | No Group | 3.90 (1.27) |
| | | Black | 3.95 (1.18) |
| | | Poor | 4.22 (1.09) |
| | | Working | 4.05 (1.13) |
| Support for communities impacted by flooding | White | Control | 4.15 (0.78) |
| | | No Group | 4.04 (0.95) |
| | | Black | 3.85 (0.97) |
| | | Poor | 4.25 (0.79) |
| | | Working | 4.23 (0.81) |
| | Black | Control | 4.39 (0.68) |
| | | No Group | 4.48 (0.59) |
| | | Black | 4.50 (0.61) |
| | | Poor | 4.51 (0.63) |
| | | Working | 4.31 (0.71) |
| Support for communities impacted by climate change | White | Control | 3.54 (1.22) |
| | | No Group | 3.73 (1.26) |
| | | Black | 3.55 (1.31) |
| | | Poor | 3.95 (1.14) |
| | | Working | 3.93 (1.12) |
| | Black | Control | 4.22 (0.77) |

| | | | |
|---|---|---|---|
| | | No Group | 4.36 (0.74) |
| | | Black | 4.42 (0.77) |
| | | Poor | 4.37 (0.77) |
| | | Working | 4.25 (0.75) |
| Flooding policy support | White | Control | 4.08 (0.73) |
| | | No Group | 4.11 (0.80) |
| | | Black | 4.16 (0.77) |
| | | Poor | 4.29 (0.62) |
| | | Working | 4.33 (0.69) |
| | Black | Control | 4.29 (0.68) |
| | | No Group | 4.39 (0.74) |
| | | Black | 4.52 (0.56) |
| | | Poor | 4.40 (0.65) |
| | | Working | 4.26 (0.81) |
| Climate policy support | White | Control | 3.79 (1.15) |
| | | No Group | 3.73 (1.26) |
| | | Black | 3.74 (1.22) |
| | | Poor | 3.95 (1.14) |
| | | Working | 4.03 (1.12) |
| | Black | Control | 4.16 (0.79) |
| | | No Group | 4.16 (0.85) |
| | | Black | 4.28 (0.70) |
| | | Poor | 4.15 (0.76) |
| | | Working | 4.09 (0.86) |

**Table S7**

*Study 2: Main Effects for Symbolic Racism and Stimulus Conditions Compared to the Control Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Heat Wave Impact | Unequal Climate Impact | Community Support - Heat Wave | Community Support - Climate | Policy Support - Heat Wave | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| No Group vs. Control | 0.083 | 0.070 | 0.008 | 0.054 | 0.033 | 0.043 |
| | (0.076) | (0.092) | (0.072) | (0.077) | (0.067) | (0.077) |
| Black vs. Control | -0.265*** | 0.016 | -0.086 | -0.090 | -0.036 | -0.064 |
| | (0.076) | (0.092) | (0.072) | (0.077) | (0.067) | (0.077) |
| Poor vs. Control | 0.053 | 0.147 | 0.101 | 0.119 | 0.020 | 0.020 |
| | (0.077) | (0.093) | (0.072) | (0.078) | (0.067) | (0.077) |
| Working vs. Control | -0.019 | 0.059 | -0.030 | -0.053 | 0.002 | -0.069 |
| | (0.077) | (0.092) | (0.072) | (0.077) | (0.067) | (0.077) |
| Racism | -0.217*** | -0.358*** | -0.417*** | -0.494*** | -0.350*** | -0.467*** |
| | (0.020) | (0.024) | (0.019) | (0.020) | (0.017) | (0.020) |
| Constant | 4.928*** | 4.901*** | 5.111*** | 5.252*** | 5.103*** | 5.210*** |
| | (0.074) | (0.090) | (0.070) | (0.075) | (0.065) | (0.075) |
| Observations | 1,419 | 1,419 | 1,419 | 1,418 | 1,418 | 1,417 |
| $R^2$ | 0.099 | 0.140 | 0.270 | 0.310 | 0.228 | 0.286 |
| Adjusted $R^2$ | 0.095 | 0.137 | 0.267 | 0.307 | 0.225 | 0.283 |
| Residual Std. Error | 0.919 (df = 1413) | 1.109 (df = 1413) | 0.862 (df = 1413) | 0.928 (df = 1412) | 0.805 (df = 1412) | 0.923 (df = 1411) |
| F Statistic | 30.892*** (df = 5; 1413) | 46.033*** (df = 5; 1413) | 104.443*** (df = 5; 1413) | 126.649*** (df = 5; 1412) | 83.344*** (df = 5; 1412) | 112.903*** (df = 5; 1411) |

*Note*. The Control condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S8**

*Study 2: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Control Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Heat Wave Impact | Unequal Climate Impact | Community Support - Heat Wave | Community Support - Climate | Policy Support - Heat Wave | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| No Group vs. Control | 0.034 | 0.097 | 0.155 | -0.041 | 0.139 | 0.160 |
| | (0.172) | (0.210) | (0.163) | (0.176) | (0.153) | (0.175) |
| Black vs. Control | 0.535** | 0.447* | 0.298 | 0.072 | 0.160 | -0.171 |
| | (0.176) | (0.214) | (0.166) | (0.179) | (0.156) | (0.179) |
| Poor vs. Control | 0.407* | 0.396 | 0.185 | 0.072 | 0.158 | 0.006 |
| | (0.178) | (0.218) | (0.169) | (0.182) | (0.158) | (0.182) |
| Working vs. Control | 0.107 | 0.199 | 0.107 | 0.018 | 0.189 | 0.071 |
| | (0.182) | (0.222) | (0.173) | (0.186) | (0.162) | (0.185) |
| Racism | -0.127** | -0.296*** | -0.362*** | -0.487*** | -0.306*** | -0.458*** |
| | (0.040) | (0.048) | (0.037) | (0.040) | (0.035) | (0.040) |
| No Group * Racism | 0.022 | -0.009 | -0.056 | 0.037 | -0.040 | -0.045 |
| | (0.060) | (0.073) | (0.057) | (0.061) | (0.053) | (0.061) |
| Black * Racism | -0.290*** | -0.157* | -0.140* | -0.058 | -0.072 | 0.037 |
| | (0.058) | (0.071) | (0.055) | (0.059) | (0.051) | (0.059) |
| Poor * Racism | -0.133* | -0.094 | -0.033 | 0.017 | -0.052 | 0.005 |
| | (0.060) | (0.074) | (0.057) | (0.062) | (0.054) | (0.062) |
| Working * Racism | -0.050 | -0.054 | -0.052 | -0.026 | -0.070 | -0.051 |
| | (0.061) | (0.075) | (0.058) | (0.063) | (0.054) | (0.062) |
| Constant | 4.691*** | 4.740*** | 4.966*** | 5.234*** | 4.986*** | 5.187*** |
| | (0.117) | (0.142) | (0.111) | (0.119) | (0.103) | (0.119) |
| Observations | 1,419 | 1,419 | 1,419 | 1,418 | 1,418 | 1,417 |
| $R^2$ | 0.120 | 0.144 | 0.273 | 0.311 | 0.229 | 0.287 |
| Adjusted $R^2$ | 0.115 | 0.138 | 0.269 | 0.307 | 0.224 | 0.283 |
| Residual Std. Error | 0.909 (df = 1409) | 1.108 (df = 1409) | 0.861 (df = 1409) | 0.929 (df = 1408) | 0.805 (df = 1408) | 0.923 (df = 1407) |

| | | | | | | |
|---|---|---|---|---|---|---|
| F Statistic | 21.445*** (df = 9; 1409) | 26.317*** (df = 9; 1409) | 58.906*** (df = 9; 1409) | 70.606*** (df = 9; 1408) | 46.542*** (df = 9; 1408) | 62.969*** (df = 9; 1407) |

*Note*. The Control condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S9**

*Study 2: Interaction Between Symbolic Racism and Experimental Conditions Compared to the No Group Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Heat Wave Impact | Unequal Climate Impact | Community Support - Heat Wave | Community Support - Climate | Policy Support - Heat Wave | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. No Group | -0.034 | -0.097 | -0.155 | 0.041 | -0.139 | -0.160 |
| | (0.172) | (0.210) | (0.163) | (0.176) | (0.153) | (0.175) |
| Black vs. No Group | 0.501** | 0.351 | 0.143 | 0.113 | 0.021 | -0.331 |
| | (0.183) | (0.223) | (0.173) | (0.187) | (0.162) | (0.185) |
| Poor vs. No Group | 0.372* | 0.299 | 0.030 | 0.112 | 0.019 | -0.153 |
| | (0.185) | (0.226) | (0.175) | (0.189) | (0.164) | (0.188) |
| Working vs. No Group | 0.073 | 0.103 | -0.048 | 0.059 | 0.050 | -0.089 |
| | (0.189) | (0.230) | (0.179) | (0.193) | (0.168) | (0.192) |
| Racism | -0.106* | -0.305*** | -0.418*** | -0.450*** | -0.346*** | -0.503*** |
| | (0.045) | (0.055) | (0.042) | (0.046) | (0.040) | (0.046) |
| Control * Racism | -0.022 | 0.009 | 0.056 | -0.037 | 0.040 | 0.045 |
| | (0.060) | (0.073) | (0.057) | (0.061) | (0.053) | (0.061) |
| Black * Racism | -0.311*** | -0.148* | -0.084 | -0.095 | -0.032 | 0.083 |
| | (0.062) | (0.075) | (0.058) | (0.063) | (0.055) | (0.063) |
| Poor * Racism | -0.155* | -0.085 | 0.023 | -0.020 | -0.012 | 0.050 |
| | (0.064) | (0.078) | (0.061) | (0.066) | (0.057) | (0.065) |
| Working * Racism | -0.071 | -0.045 | 0.004 | -0.063 | -0.030 | -0.006 |
| | (0.065) | (0.079) | (0.061) | (0.066) | (0.058) | (0.066) |
| Constant | 4.726*** | 4.837*** | 5.121*** | 5.193*** | 5.125*** | 5.347*** |
| | (0.127) | (0.155) | (0.120) | (0.130) | (0.112) | (0.129) |
| Observations | 1,419 | 1,419 | 1,419 | 1,418 | 1,418 | 1,417 |
| $R^2$ | 0.120 | 0.144 | 0.273 | 0.311 | 0.229 | 0.287 |
| Adjusted $R^2$ | 0.115 | 0.138 | 0.269 | 0.307 | 0.224 | 0.283 |
| Residual Std. Error | 0.909 (df = 1409) | 1.108 (df = 1409) | 0.861 (df = 1409) | 0.929 (df = 1408) | 0.805 (df = 1408) | 0.923 (df = 1407) |
| F Statistic | 21.445*** (df = 9; 1409) | 26.317*** (df = 9; 1409) | 58.906*** (df = 9; 1409) | 70.606*** (df = 9; 1408) | 46.542*** (df = 9; 1408) | 62.969*** (df = 9; 1407) |

*Note.* The No Group condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S10**

*Study 2: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Black Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Heat Wave Impact | Unequal Climate Impact | Community Support - Heat Wave | Community Support - Climate | Policy Support - Heat Wave | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. Black | -0.535** | -0.447* | -0.298 | -0.072 | -0.160 | 0.171 |
| | (0.176) | (0.214) | (0.166) | (0.179) | (0.156) | (0.179) |
| No Group vs. Black | -0.501** | -0.351 | -0.143 | -0.113 | -0.021 | 0.331 |
| | (0.183) | (0.223) | (0.173) | (0.187) | (0.162) | (0.185) |
| Poor vs. Black | -0.128 | -0.051 | -0.113 | -0.001 | -0.002 | 0.178 |
| | (0.188) | (0.230) | (0.178) | (0.193) | (0.167) | (0.192) |
| Working vs. Black | -0.428* | -0.248 | -0.191 | -0.054 | 0.029 | 0.242 |
| | (0.192) | (0.234) | (0.182) | (0.196) | (0.171) | (0.195) |
| Racism | -0.417*** | -0.453*** | -0.502*** | -0.545*** | -0.378*** | -0.420*** |
| | (0.042) | (0.052) | (0.040) | (0.043) | (0.038) | (0.043) |
| Control * Racism | 0.290*** | 0.157* | 0.140* | 0.058 | 0.072 | -0.037 |
| | (0.058) | (0.071) | (0.055) | (0.059) | (0.051) | (0.059) |
| No Group * Racism | 0.311*** | 0.148* | 0.084 | 0.095 | 0.032 | -0.083 |
| | (0.062) | (0.075) | (0.058) | (0.063) | (0.055) | (0.063) |
| Poor * Racism | 0.156* | 0.063 | 0.107 | 0.075 | 0.020 | -0.033 |
| | (0.062) | (0.076) | (0.059) | (0.064) | (0.055) | (0.064) |
| Working * Racism | 0.240*** | 0.103 | 0.088 | 0.032 | 0.003 | -0.089 |
| | (0.063) | (0.077) | (0.060) | (0.065) | (0.056) | (0.064) |
| Constant | 5.226*** | 5.188*** | 5.264*** | 5.306*** | 5.146*** | 5.016*** |
| | (0.131) | (0.160) | (0.124) | (0.134) | (0.116) | (0.133) |
| Observations | 1,419 | 1,419 | 1,419 | 1,418 | 1,418 | 1,417 |
| $R^2$ | 0.120 | 0.144 | 0.273 | 0.311 | 0.229 | 0.287 |
| Adjusted $R^2$ | 0.115 | 0.138 | 0.269 | 0.307 | 0.224 | 0.283 |
| Residual Std. Error | 0.909 (df = 1409) | 1.108 (df = 1409) | 0.861 (df = 1409) | 0.929 (df = 1408) | 0.805 (df = 1408) | 0.923 (df = 1407) |
| F Statistic | 21.445*** (df = 9; 1409) | 26.317*** (df = 9; 1409) | 58.906*** (df = 9; 1409) | 70.606*** (df = 9; 1408) | 46.542*** (df = 9; 1408) | 62.969*** (df = 9; 1407) |

*Note*. The Black condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001

**Table S11**

*Study 2: Interaction Between Symbolic Racism and Experimental Conditions Compared to the Poor Condition*

| | Dependent variable: | | | | | |
|---|---|---|---|---|---|---|
| | Unequal Heat Wave Impact | Unequal Climate Impact | Community Support - Heat Wave | Community Support - Climate | Policy Support - Heat Wave | Policy Support - Climate |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Control vs. Poor | -0.407* | -0.396 | -0.185 | -0.072 | -0.158 | -0.006 |
| | (0.178) | (0.218) | (0.169) | (0.182) | (0.158) | (0.182) |
| No Group vs. Poor | -0.372* | -0.299 | -0.030 | -0.112 | -0.019 | 0.153 |
| | (0.185) | (0.226) | (0.175) | (0.189) | (0.164) | (0.188) |
| Black vs. Poor | 0.128 | 0.051 | 0.113 | 0.001 | 0.002 | -0.178 |
| | (0.188) | (0.230) | (0.178) | (0.193) | (0.167) | (0.192) |
| Working vs. Poor | -0.299 | -0.197 | -0.078 | -0.053 | 0.031 | 0.064 |
| | (0.194) | (0.237) | (0.184) | (0.199) | (0.173) | (0.198) |
| Racism | -0.261*** | -0.390*** | -0.395*** | -0.470*** | -0.358*** | -0.453*** |
| | (0.046) | (0.056) | (0.043) | (0.047) | (0.041) | (0.047) |
| Control * Racism | 0.133* | 0.094 | 0.033 | -0.017 | 0.052 | -0.005 |
| | (0.060) | (0.074) | (0.057) | (0.062) | (0.054) | (0.062) |
| No Group * Racism | 0.155* | 0.085 | -0.023 | 0.020 | 0.012 | -0.050 |
| | (0.064) | (0.078) | (0.061) | (0.066) | (0.057) | (0.065) |
| Black * Racism | -0.156* | -0.063 | -0.107 | -0.075 | -0.020 | 0.033 |
| | (0.062) | (0.076) | (0.059) | (0.064) | (0.055) | (0.064) |
| Working * Racism | 0.083 | 0.040 | -0.019 | -0.043 | -0.018 | -0.056 |
| | (0.065) | (0.080) | (0.062) | (0.067) | (0.058) | (0.067) |
| Constant | 5.098*** | 5.136*** | 5.151*** | 5.306*** | 5.144*** | 5.194*** |
| | (0.135) | (0.165) | (0.128) | (0.138) | (0.120) | (0.138) |
| Observations | 1,419 | 1,419 | 1,419 | 1,418 | 1,418 | 1,417 |
| $R^2$ | 0.120 | 0.144 | 0.273 | 0.311 | 0.229 | 0.287 |
| Adjusted $R^2$ | 0.115 | 0.138 | 0.269 | 0.307 | 0.224 | 0.283 |
| Residual Std. Error | 0.909 (df = 1409) | 1.108 (df = 1409) | 0.861 (df = 1409) | 0.929 (df = 1408) | 0.805 (df = 1408) | 0.923 (df = 1407) |

| | | | | | | |
|---|---|---|---|---|---|---|
| F Statistic | 21.445*** (df = 9; 1409) | 26.317*** (df = 9; 1409) | 58.906*** (df = 9; 1409) | 70.606*** (df = 9; 1408) | 46.542*** (df = 9; 1408) | 62.969*** (df = 9; 1407) |

*Note.* The Poor condition is the reference group for comparison. *p<0.05; **p<0.01; ***p<0.001